

Abstract Algebra

HHH

July 9, 2026

Contents

I	Group Theory	1
I	Group	2
§1	Basic Definition	2
§1.1	Basic Definition	2
	Order	3
§1.2	Subgroup	4
	Subgroup generated by sets	4
	Normal Subgroup and Normalizer	4
	Center and Centralizers	5
§1.3	Coset	6
§1.4	Cyclic Groups	7
§2	Homomorphisms	8
§2.1	Homomorphisms	8
	Natural Projection	8
§2.2	Isomorphism Theorem	9
§2.3	Automorphism	10
II	Group Action	11
§1	Basic Definition	11
§1.1	Orbits and Stabilizer	12
§1.2	The Class Equation	13
§2	Group Actions by Left multiplication	13
§2.1	Left regular action	13
§2.2	G acts on (left) coset space by (left) multiplication	14
§3	Group Actions by Conjugation	15
§3.1	G acts on itself by conjugation	15
§3.2	G acts on $\mathcal{P}(G)$ by conjugation	15
§4	Sylow's Theorem	16
§4.1		18

III Symmetric Group	20
§1 Basic Definition	20
§2 The Alternating Group	21
IV Group Series	23
§1 Nilpotent Group	23
§2 Solvable Group	24
§3 Normal and Subnormal Series	26
V Structure of Groups	29
§1 Free Groups	29
§1.1 Words on Free Group	29
§1.2 Presentations	29
§2 Product and Coproduct	30
§2.1 Direct Product	30
§2.2 Free Product	31
§2.3 Weak Direct Product	31
§3 The Krull-Schmidt Theorem	32
§4 The Fundamental Theorem of Finitely Generated Abelian Groups	33
II Ring Theory	34
VI Ring Theory	35
§1 Basic Definition	35
§2 Ideal	36
§2.1 Definition and Quotient Ring	36
§2.2 Ideals generated by a set	37
§2.3 Prime ideal	38
§2.4 Maximal ideal	39
§2.5 Chinese Remainder Theorem	39
§3 Homomorphisms	40
§4 Rings of Polynomial and Formal Power Series	41
VII Factorization in Integral Domains	42
§1 Divisor Decomposition	42
§1.1 Basic definition	42
§2 $E.D \Rightarrow P.I.D \Rightarrow U.F.D$	43
§2.1 Unique Factorization Domain	43
§2.2 Principal rings and principal domains	44
§2.3 Euclidean Ring and Euclidean domain	44
§3 Factorization in Polynomial Rings	45

§3.1	Over U.F.D	46
III	Modules Theory	49
VII	Modules	50
§1	Basic Definition	50
§1.1	Submodule generated by sets	51
§1.2	Quotient module and homomorphism	51
§1.3	Annihilator	53
§2	Modules Category	53
§2.1	Direct Products and Direct Sums	53
§2.2	Free Modules	55
	Dimension and invariant dimension property	56
	Proof of invariant dimension property	56
§2.3	Vector Space	59
§2.4	Pullbacks and Pushout	60
§3	Tensor Products	61
§3.1	Basic definition	61
§3.2	Operation of tensor products	63
§3.3		64
§4	Algebra	66
§5	Modules over Principal Ideal Domains	66
§5.1	Preparatory Lemmas	67
§5.2		67
§5.3	Torsion module decomposition	68
§5.4		69
IV	Field and Galois Theory	70
IX	Field Theory	71
§1	Field Extension	71
§1.1	Basic Definition	71
§1.2	Composition fields	71
§2	Extension tower	72
§3	Generation	73
§3.1	Finitely Generated Extensions	74
§3.2	Simple Extension	74
§3.3	n -extension	75
§4	Algebraic Extension	75

§4.1	Splitting Fields	77
§4.2	Normal Extension	78
§4.3	Separable Extension	79
§4.4	Purely Inseparable Extension (char p)	79
§4.5	Separable degree	80
§4.6		81
X	Galois Theory	82
§1	Basic Definition	82
§2		83
§3	Fundamental Theorem	83
§3.1	Stable Intermediate Fields	85
§3.2	Finite Galois correspondence	86
§3.3		87
§3.4	Question	87
§4	Galois Groups	88
§4.1		88
§5	Finite Fields	89
§5.1	Extension over finite fields	90

Part I

Group Theory

Chapter I

Group

§1 Basic Definition

§1.1 Basic Definition

Definition 1.1. Let G be a set.

1. A **binary operation** $\cdot$ on G is a function $\cdot : G \times G \rightarrow G$. For any $a, b \in G$ we shall write $a \cdot b$ for $\cdot(a, b)$.
2. A binary operation $\cdot$ on G is **associative** if for all $a, b, c \in G$ we have $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
3. If $\cdot$ is a binary operation on G we say elements a and b of G **commute** if $a \cdot b = b \cdot a$. We say $\cdot$ is **commutative** if for all $a, b \in G$, $a \cdot b = b \cdot a$.

Definition 1.2. A **semigroup** is a nonempty set G together with a binary operation on G which is associative

A **monoid** is a semigroup G which contains a identity element $e \in G$ such that $ge = eg = g$ for all $g \in G$.

A **group** is a monoid G such that for every $g \in G$ there exists a inverse element $g^{-1} \in G$ such that $g^{-1}g = gg^{-1} = e$.

A semigroup G is said to be abelian or commutative if its binary operation is (iv) commutative: $ab = ba$ for all $a, b \in G$.

The **order** of a group G is the cardinal number $|G|$. G is said to be finite [resp. infinite] if $|G|$ is finite [resp. infinite].

Proposition 1.3. If G is a semigroup, then

1. for any $a_1, a_2, \dots, a_n \in G$ the value of $a_1 \cdot a_2 \cdot \dots \cdot a_n$ is independent of how the expression is bracketed (this is called the generalized associative law).
2. $x^{a+b} = x^a x^b$
3. $(x^a)^b = x^{ab}$

If G is a monoid,

4. the identity of G is unique

If G is a group,

5. for each $a \in G$, a^{-1} is unique
6. $(a^{-1})^{-1} = a$ for all $a \in G$
7. $(a \cdot b)^{-1} = (b^{-1}) \cdot (a^{-1})$
8. a^{-n} is defined to be $(a^{-1})^n = (a^n)^{-1}$
9. for all $a, b, c \in G$, $ab = ac \Rightarrow b = c$ and $ba = ca \Rightarrow b = c$ (left and right. cancellation);
10. for $a, b \in G$ the equations $ax = b$ and $ya = b$ have unique solutions in G : $x = a^{-1}b$ and $y = ba^{-1}$.

Proposition 1.4. Let G be a semigroup. Then the following conditions on G are equivalent

1. G is a group
2. There exists an element $e \in G$ such that $ea = a$ for all $a \in G$ (left identity element); for each $a \in G$, there exists an element $a^{-1} \in G$ such that $a^{-1}a = e$ (left inverse).
3. There exists an element $e \in G$ such that $ae = a$ for all $a \in G$ (right identity element); for each $a \in G$, there exists an element $a^{-1} \in G$ such that $aa^{-1} = e$ (right inverse).
4. For all $a, b \in G$ the equations $ax = b$ and $yb = a$ have solutions in G .

Theorem 1.5. Let $\sim$ be an equivalence relation on a monoid G . Then the set $G/\sim$ is a monoid under the binary operation defined by $\bar{a} \cdot \bar{b} = \overline{ab}$,

If G is an [abelian] group, then so is G/R .

An equivalence relation on a monoid G that satisfies the hypothesis of the theorem is called a congruence relation on G .

Order

Definition 1.6. For G a group and $x \in G$ define the **order** of x to be the smallest positive integer n such that $x^n = 1$, and denote this integer by $|x|$. In this case x is said to be of order n . If no positive power of x is the identity, the order of x is defined to be infinity and x is said to be of **infinite order**.

Theorem 1.7. If x and g are elements of the group G , then

- (1) $|x| = |g^{-1}xg|$
- (2) $|ab| = |ba|$ for all $a, b \in G$
- (3) If $x^n = 1$ and $x^m = 1$, then $x^d = 1$, where $d = (m, n)$.
- (4) If $x^m = 1$ for some $m \in \mathbb{Z}$, then $|x|$ divides m .

§1.2 Subgroup

Definition 1.8. Let G be a group. The subset H of G is a **subgroup** of G if H is nonempty and H is closed under products and inverses (i.e., $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$). If H is a subgroup of G we shall write $H \leq G$.

Proposition 1.9 (The subgroup criterion). Let G be a group, then

1. A subset H of a group G is a subgroup if and only if
 - (i) $H \neq \emptyset$, and
 - (ii) for all $x, y \in H$, $xy^{-1} \in H$. Furthermore, if H is finite, then it suffices to check that H is nonempty and closed under multiplication.
2. If $\mathcal{A}$ is any nonempty collection of subgroups of G , then the intersection of all members of $\mathcal{A}$ is also a subgroup of G .
3. If $\mathcal{B}$ is a chain (with respect to set inclusion) in the family of all subgroups of G , then the union of all members in $\mathcal{B}$ is also a subgroup of G .

Proposition 1.10. Let H and K be subgroups of G . Then $H \cup K$ is a subgroup if and only if either $H \subseteq K$ or $K \subseteq H$.

Subgroup generated by sets

Definition 1.11. If X is any subset of the group G define

$$\langle X \rangle = \bigcap_{X \subseteq H \leq G} H$$

This is called the **subgroup of G generated by X** . When X is the finite set $\{a_1, a_2, \dots, a_n\}$ we write $\langle a_1, a_2, \dots, a_n \rangle$ for the group generated by $a_1, a_2, \dots, a_n$. If A and B are two subsets of G we shall write $\langle A, B \rangle$ in place of $\langle A \cup B \rangle$.

If H and K are subgroups, $\langle H \cup K \rangle$ is called the **join** of H and K and is denoted $H \vee K$ (additive notation: $H + K$)

Theorem 1.12. Suppose a group G and subset $A \subset G$, then

$$\langle A \rangle = \{a_1^{\epsilon_1} a_2^{\epsilon_2} \dots a_n^{\epsilon_n} \mid n \in \mathbb{Z}, n \geq 0 \text{ and } a_i \in A, \epsilon_i = \pm 1 \text{ for each } i\}$$

$$(\langle A \rangle = \{e\} \text{ if } A = \emptyset)$$

Normal Subgroup and Normalizer

Definition 1.13. Let G be a group.

1. A subgroup N of a group G is called **normal subgroup** if every element of G normalizes N , i.e. $gNg^{-1} = N$ for all $g \in G$. If N is a normal subgroup of G we shall write $N \trianglelefteq G$.
2. Define the **normalizer of X in G** to be the set (subgroup)

$$N_G(X) = \{g \in G \mid gXg^{-1} = X\}$$

If K is any subset of $N_G(X)$, we shall say K **normalizes** X .

Theorem 1.14. Let N be a subgroup of the group G . The following conditions are equivalent:

1. $N \trianglelefteq G$
2. $N_G(N) = G$
3. $gN = Ng$ for all $g \in G$
4. $gNg^{-1} \subseteq N$ for all $g \in G$.
5. Left and right congruence modulo N coincide (that is, define the same equivalence relation on G);
6. every left coset of N in G is a right coset of N in G
7. for any $x, y \in G$, $xy \in N \Leftrightarrow yx \in N$.

Theorem 1.15. Let K and N be subgroups of a group G with N normal in G . Then

- (1) $N \cap K$ is a normal subgroup of K ;
- (2) N is a normal subgroup of $N \vee K$;
- (3) $NK = N \vee K = KN$;
- (4) if K is normal in G and $K \cap N = \langle e \rangle$, then $nk = kn$ for all $k \in K$ and $n \in N$.

Definition 1.16. If N is a normal subgroup of a group G and G/N is the set of all (left) cosets of N in G , then G/N is a group of order $[G : N]$ under the binary operation given by $(aN) \cdot (bN) = abN$. Then the group G/N is called the **quotient group of G by N** .

Center and Centralizers

Definition 1.17. Let X be any nonempty subset of G . Define

$$C_G(X) = \{g \in G : gxg^{-1} = x \text{ for all } x \in X\}$$

This subset of G is called the **centralizer of X in G** . $C_G(X)$ is a subgroup of G which consists of element commute with every element of X . Define

$$Z(G) = C_G(G) = \{g \in G : gag^{-1} = a \text{ for all } a \in G\}$$

the set of elements commuting with all the elements of G . This subset of G is called the **center of G** .

Proposition 1.18. Suppose a group G .

1. $C_G(Z(G)) = G$
2. $N_G(Z(G)) = G$
3. If A and B are subsets of G with $A \subseteq B$ then $C_G(B) \leq C_G(A)$.

§1.3 Coset

Definition 1.19. For any $H \leq G$ and any $g \in G$ let

$$gH = \{gn : n \in H\} \quad \text{and} \quad Hg = \{ng : n \in H\}$$

called respectively a **left coset** and a **right coset of H in G** . Any element of a coset is called a **representative for the coset**.

Theorem 1.20. Let H be any subgroup of the group G . Then

1. The set $\{gH : g \in G\}$ of left cosets of H in G form a partition of G .
2. For all $u, v \in G$, $uH = vH$ if and only if $v^{-1}u \in H$, denoted

$$u \equiv_l v \pmod{H}$$

This is a congruence relation.

3. Right [resp. left] congruence modulo H is an equivalence relation on G

Theorem 1.21. If K, H, G are groups with $K < H < G$, then $[G : K] = [G : H][H : K]$. If any two of these indices are finite, then so is the third.

Theorem 1.22 (Lagrange's Theorem). If G is a finite group and H is a subgroup of G , then the order of H divides the order of G and the number of left cosets of H in G equals $\frac{|G|}{|H|}$ called the **index of H in G** and is denoted by $[G : H]$.

Corollary 1.23. If G is a finite group and $x \in G$, then the order of x divides the order of G . In particular $x^{|G|} = 1$ for all x in G .

Corollary 1.24. If G is a group of prime order p , then G is cyclic, hence $G \cong Z_p$.

Definition 1.25. Let H and K be subsets of a group and define

$$HK = \{hk \mid h \in H, k \in K\}.$$

Theorem 1.26. *If H and K are finite subgroups of a group then*

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proposition 1.27. *If H and K are subgroups of a group, HK is a subgroup if and only if $HK = KH$.*

Corollary 1.28. *If H and K are subgroups of G and $H \leq N_G(K)$, then HK is a subgroup of G . In particular, if $K \trianglelefteq G$ then $HK = KH \leq G$ for any $H \leq G$.*

Definition 1.29. *The **quotient group**, G/N (read G modulo N), is*

$$\{gN : g \in G\}$$

with the operation defined by

$$g_1N \cdot g_2N = g_1Ng_2N = g_1g_2N$$

Proposition 1.30. (1) *If $N \leq Z(G)$, then $N \trianglelefteq G$.*

(2) $Z(G) \trianglelefteq G$.

§1.4 Cyclic Groups

Definition 1.31. *A group G is **cyclic** if H can be generated by a single element, i.e., there is some element $x \in H$ such that $G = \{x^n : n \in \mathbb{Z}\}$.*

Theorem 1.32. *Any two cyclic groups of the same order are isomorphic. More specifically,*

(1) *If $n \in \mathbb{Z}_{>0}$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n , then the map*

$$\begin{aligned} \varphi : \langle x \rangle &\rightarrow \langle y \rangle \\ x^k &\mapsto y^k \end{aligned}$$

is well defined and is an isomorphism.

(2) *If $\langle x \rangle$ is an infinite cyclic group, the map*

$$\begin{aligned} \varphi : \mathbb{Z} &\rightarrow \langle x \rangle \\ k &\mapsto x^k \end{aligned}$$

is well defined and is an isomorphism.

§2 Homomorphisms

§2.1 Homomorphisms

Definition 2.1. Let $(G, \cdot)$ and $(H, \times)$ be groups. A map $\varphi : G \rightarrow H$ such that

$$\varphi(x \cdot y) = \varphi(x) \times \varphi(y), \quad \text{for all } x, y \in G$$

is called a **group homomorphism**.

If f is injective as a map of sets, f is said to be a **monomorphism**. If f is surjective, f is called an **epimorphism**. If f is bijective, f is called an **isomorphism**. In this case G and H are said to be **isomorphic** (written $G \cong H$). A homomorphism $f : G \rightarrow G$ is called an **endomorphism** of G and an isomorphism $f : G \rightarrow G$ is called an **automorphism** of G .

Definition 2.2. Let f is a homomorphism $f : G \rightarrow H$. The **kernel** of homomorphism f is the set

$$\{g \in G \mid f(g) = e\}$$

and will be denoted by $\text{Ker } f$.

Theorem 2.3. Let $f : G \rightarrow H$ be a homomorphism of groups. Then

- (1) f is a monomorphism if and only if $\text{Ker } f = e$;
- (2) f is an isomorphism if and only if there is a homomorphism $f^{-1} : H \rightarrow G$ such that $f f^{-1} = 1_H$ and $f^{-1} f = 1_G$.

Proposition 2.4. Let G and H be groups and let $\varphi : G \rightarrow H$ be a homomorphism.

1. $\varphi(1_G) = 1_H$, where 1_G and 1_H are the identities of G and H , respectively.
2. $\varphi(g^n) = \varphi(g)^n$ for all $n \in \mathbb{Z}$.
3. $\ker \varphi \trianglelefteq G$.
4. $\text{Im}(\varphi)$ is a subgroup of H .

Natural Projection

Definition 2.5. Let $N \trianglelefteq G$. The homomorphism

$$\pi : G \rightarrow G/N$$

defined by

$$\pi(g) = gN = Ng$$

is called the **natural projection** (homomorphism) of G onto G/N . If $\bar{H} \leq G/N$ is a subgroup of G/N , the **complete preimage** of $\bar{H}$ in G is the preimage of $\bar{H}$ under the natural projection homomorphism.

Theorem 2.6. *A subgroup N of the group G is normal if and only if it is the kernel of some homomorphism.*

§2.2 Isomorphism Theorem

Theorem 2.7 (The First Isomorphism Theorem). *If $\varphi : G \rightarrow H$ is a homomorphism of groups, then*

$$G / \ker \varphi \cong \varphi(G)$$

and

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & \varphi(G) \\ \pi \downarrow & \nearrow & \\ G / \ker \varphi & & \end{array}$$

Corollary 2.8. *Let $\varphi : G \rightarrow H$ be a homomorphism of groups.*

1. φ is injective if and only if $\ker \varphi = 1$.
2. $|G : \ker \varphi| = |\varphi(G)|$.

Proposition 2.9. *If H and K are subgroups of a group, HK is a subgroup if and only if $HK = KH$.*

Proposition 2.10. *Let G be a group, let A and B be subgroups of G and assume $A \leq N_G(B)$ (particularly, $B \trianglelefteq A$ or $A, B \trianglelefteq G$). Then*

- (1) $B \trianglelefteq AB = BA \leq G$
- (2) $A \cap B \trianglelefteq A$

Theorem 2.11 (The Second Isomorphism Theorem). *Let G be a group, let A and B be subgroups of G and assume $A \leq N_G(B)$. Then*

$$AB/B \cong A/A \cap B$$

Theorem 2.12 (The Third Isomorphism Theorem). *Let G be a group and $H \trianglelefteq G$. Then for each $H \leq K \trianglelefteq G$ we have $K/H \trianglelefteq G/H$ and*

$$(G/H)/(K/H) \cong G/K$$

If we denote the quotient by H with a bar, this can be written

$$\bar{G}/\bar{K} \cong G/K$$

Theorem 2.13 (The Fourth or Lattice Isomorphism Theorem). *Let G be a group and let N be a normal subgroup of G . Then there is a bijection from the set $\mathcal{U} = \{H : N < H < G\}$ onto the set*

$\mathcal{V} = \{\bar{H} : \bar{H} < \bar{G}\}$. This bijection has the following properties: for all $A, B \leq G$ with $N \leq A$ and $N \leq B$

1. $A \leq B$ if and only if $\bar{A} \leq \bar{B}$,
2. if $A \leq B$, then $|B : A| = |\bar{B} : \bar{A}|$,
3. $\overline{\langle A, B \rangle} = \langle \bar{A}, \bar{B} \rangle$,
4. $\overline{A \cap B} = \bar{A} \cap \bar{B}$
5. $A \trianglelefteq G$ if and only if $\bar{A} \trianglelefteq \bar{G}$.

§2.3 Automorphism

Definition 2.14. Let G be a group. An isomorphism from G onto itself is called an **automorphism** of G . The set of all automorphisms of G is denoted by $\text{Aut}(G)$.

$\text{Aut}(G)$ is a subgroup of S_G .

Proposition 2.15. Let H be a normal subgroup of the group G . Then G acts by conjugation on H as automorphisms of H

$$\sigma_g : h \mapsto ghg^{-1} \quad \text{for each } h \in H$$

For each $g \in G$, conjugation by g is an automorphism of H .

The permutation representation $\sigma : g \mapsto \sigma_g$ afforded by this action is a homomorphism of G into $\text{Aut}(H)$ with kernel $C_G(H)$ that

$$G/C_G(H) \cong \sigma(G) \leq \text{Aut}(H)$$

In particular, $G/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$.

Corollary 2.16. If K is any subgroup of the group G and $g \in G$, then $K \cong gKg^{-1}$. Conjugate elements and conjugate subgroups have the same order.

Corollary 2.17. For $H \leq G$, the quotient group $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$. (Observe that $H \trianglelefteq N_G(H) = G'$ and $C_{G'}(H) = C_G(H)$)

In particular, $G/Z(G)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Definition 2.18. Let G be a group and let $g \in G$. Conjugation by g is called an **inner automorphism** of G and the subgroup of $\text{Aut}(G)$ consisting of all inner automorphisms is denoted by $\text{Inn}(G)$.

$$G/Z(G) \cong \text{Inn}(G) \leq \text{Aut}(H)$$

Definition 2.19. A subgroup H of a group G is called **characteristic** in G , denoted $H \text{ char } G$, if every automorphism of G maps H to itself, i.e., $\sigma(H) = H$ for all $\sigma \in \text{Aut}(G)$.

Chapter II

Group Action

§1 Basic Definition

Definition 1.1. A **group action** of a group G on a set X is a map from $G \times X$ to X (written as $g \cdot x$, for all $g \in G$ and $x \in X$) satisfying the following properties:

$$(i) \quad g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x, \text{ for all } g_1, g_2 \in G, x \in X$$

$$(ii) \quad e \cdot x = x \text{ for all } x \in X.$$

If G acts on a set X and distinct elements of X induce distinct permutations of X , the action is said to be **faithful**.

The **kernel of the action** of G on X is defined as

$$\{g \in G : g \cdot x = x, \text{ for all } x \in X\}$$

Proposition 1.2. Let G acts on X , then

1. for each fixed $g \in G$

$$\sigma_g : x \mapsto g \cdot x$$

is a permutation of X .

2. the map

$$\sigma : G \rightarrow \mathfrak{S}_X \quad \text{provided that } g \mapsto \sigma_g$$

is a homomorphism called the **permutation representation associated to the action**.

3. Conversely, if $\varphi : G \rightarrow \mathfrak{S}_X$ is any homomorphism, then the map from $G \times X$ to X defined by

$$g \cdot a = \varphi(g)(a) \quad \text{for all } g \in G, \text{ and all } a \in X$$

satisfies the properties of a group action of G on X .

4. Thus the action of G on X and $\text{Hom}(G, \mathfrak{S}_X)$ are in bijective correspondence. And the kernel of an action of the group G on the set X is the same as the kernel of the corresponding permutation representation $\sigma : G \rightarrow \mathfrak{S}_X$.

Corollary 1.3. *Let G acts on X , then*

1. *the action is faithful if and only if the associated permutation representation is injective.*
2. *the action is faithful if and only if the kernel of the action is $\{e\}$*
3. *if G be a group acting on X and K be the kernel of the action, then G/K acts on X by*

$$(g + K) \cdot x = g \cdot x$$

faithfully

§1.1 Orbits and Stabilizer

Definition 1.4. *Let G be a group acting on the nonempty set X .*

1. *The equivalence class*

$$\mathcal{O}_x = \{g \cdot x : g \in G\}$$

*is called the **orbit** of G containing x .*

2. *If $|\mathcal{O}_x| = 1$, i.e. $g \cdot x = x$ for all $g \in G$, then we call x **fixed element** of X .*
3. *The action of G on X is called **transitive** if there is only one orbit of G .*

Definition 1.5. *If G is a group acting on a set X and x is some fixed element of X , the **stabilizer of x in G** is the subgroup*

$$G_x = \{g \in G \mid g \cdot x = x\}$$

Proposition 1.6. *Let G act on the set X and $x, y \in X$*

1. *The kernel of the action is $\bigcap_{x \in X} G_x$*
2. *If $y = g \cdot x$, then $G_y = gG_xg^{-1}$.*
3. *Thus if G acts transitively on X then the kernel of the action is*

$$\bigcap_{g \in G} gG_xg^{-1}$$

where x is any element in X .

§1.2 The Class Equation

Definition 1.7. Let G be a group acts on X and X' respectively. If there exists a one-to-one map φ that

$$\varphi(g \cdot x) = g \cdot \varphi(x)$$

for all $g \in G$ and $x \in X$. We say the two action are **equivalent**.

Theorem 1.8. Let G be a group acting on the nonempty set X , $x \in X$. Then the action on $\{gG_x : g \in G\}$ by left multiplication and the action of G on $\mathcal{O}_x$ are equivalent.

Proof. We define

$$\psi : \{gG_x : g \in G\} \longrightarrow \mathcal{O}_x \text{ given that } gG_x \mapsto g \cdot x$$

Then for any $a \in G$,

$$\psi(a \cdot gG_x) = \psi(agG_x) = ag \cdot x = a \cdot (g \cdot x) = a \cdot \psi(G_x)$$

□

Corollary 1.9 (The Class Equation). Let G be group acting on a finite X . Then

1. The number of elements in the orbit of $x \in G$ is

$$\#\mathcal{O}_x = [G : G_x]$$

2. let $x_1, x_2, \dots, x_r$ be representatives of the distinct orbit of X , we have partition

$$X = \bigsqcup \mathcal{O}_{x_i}$$

Furthermore,

$$\#X = \sum_{i=1}^r [G : G_{x_i}]$$

§2 Group Actions by Left multiplication

§2.1 Left regular action

Definition 2.1. Let G be any group and acts on itself defined by

$$g \cdot x = gx$$

for each $g \in G$ and $x \in G$. This action is called the **left regular action** of G on itself. Associated to the left regular action, the homomorphism

$$\varphi : G \rightarrow \text{Aut}(G)$$

defined by

$$g \rightarrow \sigma_g$$

is called **left regular representation**.

Corollary 2.2 (Cayley's Theorem). *Every group is isomorphic to a subgroup of some symmetric group. If G is a group of order n , then G is isomorphic to a subgroup of S_n .*

§2.2 G acts on (left) coset space by (left) multiplication

Theorem 2.3. *Let G be a group, H be a subgroup of G . If G act on the set $X = \{xH : x \in G\}$ by left multiplication. Then*

1. G acts transitively on X
2. The stabilizer of $xH \in X$ is the subgroup xHx^{-1}
3. the kernel of the action π_H is

$$\bigcap_{x \in G} G_{xH} = \bigcap_{x \in G} xHx^{-1}$$

thus $\text{Ker } \pi_H$ is the largest normal subgroup of G contained in H .

4. The set of all fixed elements in X is

$$\{xH : xHx^{-1} = H\} = \{xH : x \in N_G(H)\}$$

Thus the number of all distinct fixed elements are

$$\#\{xH : x \in N_G(H)\} = [N_G(H) : H]$$

Corollary 2.4. *If H is a subgroup of index n in a group G and no nontrivial normal subgroup of G is contained in H , then G is isomorphic to a subgroup of $\mathfrak{S}_n$.*

Corollary 2.5. *If H is a subgroup of a finite group G of index p , where p is the smallest prime dividing $|G|$, then H is normal in G .*

Proof. Let X be the set of all left cosets of H in G and $\pi_H : G \rightarrow X$ be the associated permutation representation. By 2.3, $\text{Ker } \pi_H$ is normal in G and contained in H . Furthermore $G/\text{Ker } \pi_H$ is isomorphic to a subgroup of $\mathfrak{S}_X \cong \mathfrak{S}_p$ by 1.3 and ???. Hence $|G/\text{Ker } \pi_H| = [H : \text{Ker } \pi_H] [G : H]$ divides $p!$, we must have $[H : \text{Ker } \pi_H] = 1$ since $[H : \text{Ker } \pi_H]$ divide $|G|$ and the minimality of p . Thus $H = \text{Ker } \pi_H$ is normal in G . $\square$

§3 Group Actions by Conjugation

§3.1 G acts on itself by conjugation

Definition 3.1. Suppose G is any group and we consider G acting on itself by conjugation:

$$g \cdot a = gag^{-1} \quad \text{for all } g \in G, a \in G$$

Two elements a and b of G are said to be **conjugate** in G if there is some $g \in G$ such that $b = gag^{-1}$. The orbits of G acting on itself by conjugation are called the **conjugacy classes** of G .

Proposition 3.2. Let G be a finite group and act on itself by conjugation.

1. For each $g \in G$, conjugation by g induces an automorphism of G .

$$\sigma_g : x \mapsto gxg^{-1}$$

Thus there is a homomorphism $\sigma : G \rightarrow \text{Aut } G$ whose kernel is $C(G) = \{g \in G \mid gx = xg \text{ for all } x \in G\}$. The automorphism σ_g is called the **inner automorphism** induced by g .

2. The stabilizer of $s \in G$ is

$$G_s = \{g \in G : g \cdot s = gs g^{-1} = s\} = C_G(s)$$

is the centralizer of s in G .

- 3.

$$\#\mathcal{O}_s = [G : C_G(s)]$$

4. Let $g_1, g_2, \dots, g_r$ be representatives of the distinct conjugacy classes of G not contained in the center $Z(G)$ of G . Then

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(g_i)]$$

§3.2 G acts on $\mathcal{P}(G)$ by conjugation

Definition 3.3. A group G acts on the set $\mathcal{P}(G)$ of all subsets of itself by defining

$$g \cdot S = gSg^{-1}$$

for any $g \in G$ and $S \in \mathcal{P}(G)$. Two subsets S and T of G are said to be **conjugate** in G if there is some $g \in G$ such that $T = gSg^{-1}$.

Proposition 3.4. For action by conjugation,

1. The stabilizer G_S of S

$$G_S = \{g \in G : g \cdot S = gSg^{-1} = S\} = N_G(S)$$

is the normalizer of S in G .

2. The number of conjugates of a subset S in a group G is the index of the normalizer of S , that is, $\#O_S = [G : N_G(S)]$.

Corollary 3.5. *If H is any a nontrivial normal subgroup of G then $H \cap Z(G) \neq 1$. In particular, every normal subgroup of order p is contained in the center.*

§4 Sylow's Theorem

Definition 4.1. *Let G be a group and let p be a prime.*

1. *A group of order p^α for some $\alpha \geq 1$ is called a **p -group**. Subgroups of G which are p -groups are called **p -subgroups**.*
2. *If G is a group of order $p^\alpha m$, where $p \nmid m$, then a subgroup of order p^α is called a **Sylow p -subgroup** of G .*
3. *The set of Sylow p -subgroups of G will be denoted by $\text{Syl}_p(G)$ and the number of Sylow p -subgroups of G will be denoted by $n_p(G)$.*

Lemma 4.2. *If a p -group G acts on a finite set X , then the number of all fixed elements in X*

$$\#\{x \in X : \#O_x = 1\} \equiv \#X \pmod{p}$$

Corollary 4.3. *The center $C(G)$ of a nontrivial finite p -group G contains more than one element.*

Theorem 4.4 (Cauchy). *If G is a finite group whose order is divisible by a prime p , then G contains an element of order p .*

Proof. (J. H. McKay) It is equivalent that equations $x^p = e$ has at least a solution in G . Let S be the set of p -tuples of group elements $\{(a_1, a_2, \dots, a_p) \mid a_i \in G \text{ and } a_1 a_2 \cdots a_p = e\}$. Since a_p is uniquely determined as $(a_1 a_2 \cdots a_{p-1})^{-1}$, it follows that $|S| = n^{p-1}$, where $|G| = n$. Since $p|n$, $|S| \equiv 0 \pmod{p}$.

Let the group $\mathbb{Z}_p$ act on S by cyclic permutation; that is, for $k \in \mathbb{Z}_p$,

$$k(a_1, a_2, \dots, a_p) = (a_{k+1}, a_{k+2}, \dots, a_p, a_1, \dots, a_k)$$

Verify that $(a_{k+1}, a_{k+2}, \dots, a_k) \in S$; for $0, k, k' \in \mathbb{Z}_p$ and $x \in S$, $0x = x$ and $(k + k')x = k(k'x)$. Therefore the action of $\mathbb{Z}_p$ on S is well defined.

Now $(a_1, \dots, a_p) \in S_0$ if and only if $a_1 = a_2 = \dots = a_p$; clearly $(e, e, \dots, e) \in S_0$ and hence $|S_0| \neq 0$. By 4.2, $0 \equiv |S| \equiv |S_0| \pmod{p}$. Since $|S_0| \neq 0$ there must be at least p elements in S_0 ; that is, there is $a \neq e$ such that $(a, a, \dots, a) \in S_0$ and hence $a^p = e$. Since p is prime, $|a| = p$. $\square$

Corollary 4.5. *A finite group G is a p -group if and only if every element has a order of p .*

Lemma 4.6. *If H is a p -subgroup of a finite group G , then $[N_G(H) : H] \equiv [G : H] \pmod{p}$.*

Proof. Consider G acts on $X = \{xH\}$ by left multiplication. It follows from 2.3 and 4.2 that

$$[N_G(H) : H] = \#\{xH : \#\mathcal{O}_{xH} = 1\} \equiv \#X = [G : H] \pmod{p}$$

$\square$

Corollary 4.7. *If H is p -subgroup of a finite group G such that p divides $[G : H]$, then $N_G(H) \neq H$.*

Theorem 4.8 (First Sylow Theorem). *Let G be a group of order $p^n m$, with $n \geq 1$, p prime, and $(p, m) = 1$. Then G contains a subgroup of order p^i for each $1 \leq i \leq n$ and every subgroup of G of order p^i ($i < n$) is normal in some subgroup of order p^{i+1} .*

Proof. Since $p \mid |G|$, G contains an element a , and therefore, a subgroup $\langle a \rangle$ of order p by Cauchy's Theorem. Proceeding by induction assume H is a subgroup of G of order p^i ($1 \leq i < n$). Then $p \mid [G : H]$ and by Lemma 5.5 and Corollary 5.6 H is normal in $N_G(H)$, $H \neq N_G(H)$ and $1 < |N_G(H)/H| = [N_G(H) : H] \equiv [G : H] \equiv 0 \pmod{p}$. Hence $p \mid |N_G(H)/H|$ and $N_G(H)/H$ contains a subgroup of order p as above. By 1.12 this group is of the form H_1/H where H_1 is a subgroup of $N_G(H)$ containing H . Since H is normal in $N_G(H)$, H is necessarily normal in H_1 . Finally $|H_1| = |H| |H_1/H| = p^i p = p^{i+1}$. $\square$

Corollary 4.9. *Let G be a group of order $p^n m$ with p prime, $n \geq 1$ and $(m, p) = 1$.*

- (1) *H is a Sylow p -subgroup of G if and only if H is a maximal p -subgroup of G .*
- (2) *Every conjugate of a Sylow p -subgroup is a Sylow p -subgroup.*
- (3) *If there is only one Sylow p -subgroup P , then P is normal in G .*

Theorem 4.10 (Second Sylow theorem). *If H is a p -subgroup of a finite group G , and P is any Sylow p -subgroup of G , then there exists $x \in G$ such that $H < xPx^{-1}$. In particular, any two Sylow p -subgroups of G are conjugate.*

Proof. Let X be the set of left cosets of P in G and let H act on X by (left) translation.

$$\#X_0 \equiv \#X = [G : P] \pmod{p}$$

by 4.2. But $p \nmid [G : P]$; therefore $\#X_0 \neq 0$ and there exists $xP \in X_0$, that is, the stabilizer of xP in H

$$H_{xP} = xPx^{-1} \cap H = H$$

by 2.3 and 1.9. Thus $H < xPx^{-1}$. $\square$

Theorem 4.11 (Third Sylow Theorem). *If G is a finite group and p a prime, then the number of Sylow p -subgroups of G , $n_p = [G : N_G(P)]$ divides $|G|$ and $n_p \equiv 1 \pmod{p}$.*

Proof. (1) By the second Sylow Theorem the number of Sylow p -subgroups is the number of conjugates of any one of them, say P . But this number is $[G : N_G(P)]$, a divisor of $|G|$.

(2) Let X be the set of all Sylow p -subgroups of G and let P act on X by conjugation. Then the set of all fixed elements in X

$$X_0 = \{Q : gQg^{-1} = Q \text{ for all } g \in P\} = \{Q : P < N_G(Q)\}$$

Both P and Q are Sylow p -subgroups of G and hence of $N_G(Q)$ and are therefore conjugate in $N_G(Q)$. But since Q is normal in $N_G(Q)$, this can only occur if $Q = P$ by 4.10. Therefore, $X_0 = \{P\}$ and by 4.2, $n_p = \#X \equiv \#X_0 = 1 \pmod{p}$. $\square$

Corollary 4.12. *If P is a Sylow p -subgroup of a finite group G , then $N_G(N_G(P)) = N_G(P)$.*

Proof. Every conjugate of P is a Sylow p -subgroup of G and of any subgroup of G that contains it. Since P is normal in $N = N_G(P)$, P is the only Sylow p -subgroup of N by 4.9. Therefore,

$$x \in N_G(N) \Rightarrow xNx^{-1} = N \Rightarrow xPx^{-1} < N \Rightarrow xPx^{-1} = P \Rightarrow x \in N.$$

Hence $N_G(N_G(P)) < N$; the other inclusion is obvious. $\square$

§4.1

Lemma 4.13. *Lemma 19. Let $P \in \text{Syl}_p(G)$. If Q is any p -subgroup of G , then $Q \cap N_G(P) = Q \cap P$.*

Proof. Let $H = N_G(P) \cap Q$. Since $P \leq N_G(P)$ it is clear that $P \cap Q \leq H$, so we must prove the reverse inclusion. Since by definition $H \leq Q$, this is equivalent to showing $H \leq P$. We do this by demonstrating that PH is a p -subgroup of G containing both P and H ; but P is a p -subgroup of G of largest possible order, so we must have $PH = P$, i.e., $H \leq P$.

Since $H \leq N_G(P)$, by Corollary 15 in Section 3.2, PH is a subgroup. By Proposition 13 in the same section

$$|PH| = \frac{|P||H|}{|P \cap H|}.$$

All the numbers in the above quotient are powers of p , so PH is a p -group. Moreover, P is a subgroup of PH so the order of PH is divisible by p^α , the largest power of p which divides $|G|$. These two facts force $|PH| = p^\alpha = |P|$. This in turn implies $P = PH$ and $H \leq P$. This establishes the lemma. $\square$

Corollary 4.14. *Let P be a Sylow p -subgroup of G . Then the following are equivalent:*

- (1) P is the unique Sylow p -subgroup of G , i.e., $n_p = 1$
- (2) P is normal in G
- (3) P is characteristic in G

(4) All subgroups generated by elements of p -power order are p -groups, i.e., if X is any subset of G such that $|x|$ is a power of p for all $x \in X$, then $\langle X \rangle$ is a p -group.

Definition 4.15. A **maximal subgroup** of a group G is a proper subgroup M of G such that there are no subgroups H of G with $M < H < G$.

Chapter III

Symmetric Group

§1 Basic Definition

Definition 1.1. Let Ω be any nonempty set and let S_Ω be the set of all bijections from Ω to itself (i.e., the set of all permutations of Ω).

The set S_Ω is a group under function composition: $\circ$. Note that $\circ$ is a binary operation on S_Ω since if $\sigma : \Omega \rightarrow \Omega$ and $\tau : \Omega \rightarrow \Omega$ are both bijections, then $\sigma \circ \tau$ is also a bijection from Ω to Ω . Since function composition is associative in general, $\circ$ is associative. The identity of S_Ω is the permutation 1 defined by $1(a) = a$, for all $a \in \Omega$. For every permutation σ there is a (2-sided) inverse function, $\sigma^{-1} : \Omega \rightarrow \Omega$ satisfying $\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = 1$. Thus, all the group axioms hold for $(S_\Omega, \circ)$. This group is called the **symmetric group** on the set Ω .

In the special case when $\Omega = \{1, 2, 3, \dots, n\}$, the symmetric group on Ω is denoted S_n .

A **cycle** is a string of integers which represents the element of S_n which cyclically permutes these integers and fixes all other integers. The cycle $(a_1 a_2 \dots a_m)$ is the permutation which sends a_i to a_{i+1} , $1 \leq i \leq m-1$ and sends a_m to a_1 .

We can represent this description of σ by **cycle decomposition**.

The length of a cycle is the number of integers which appear in it. A cycle of length t is called a **t-cycle**. A 2-cycle is called a **transposition**.

$$(a_1 a_2 \dots a_m) = (a_1 a_m)(a_1 a_{m-1})(a_1 a_{m-2}) \dots (a_1 a_2)$$

Two cycles are called disjoint if they have no numbers in common.

Proposition 1.2. The order of an element in S_n equals the least common multiple of the lengths of the cycles in its cycle decomposition.

Proposition 1.3. Let σ, τ be elements of the symmetric group S_n and

(1) suppose σ has cycle decomposition

$$(a_1 a_2 \dots a_{k_1})(b_1 b_2 \dots b_{k_2}) \dots$$

Then $\tau\sigma\tau^{-1}$ has cycle decomposition

$$(\tau(a_1)\tau(a_2)\dots\tau(a_{k_1}))(\tau(b_1)\tau(b_2)\dots\tau(b_{k_2}))\dots,$$

(2) Suppose σ has the form

$$\begin{pmatrix} 1 & 2 & \cdots & n \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}$$

then $\tau\sigma\tau^{-1}$ is

$$\begin{pmatrix} \tau(1) & \tau(2) & \cdots & \tau(n) \\ \tau(a_1) & \tau(a_2) & \cdots & \tau(a_n) \end{pmatrix}$$

Definition 1.4. (1) If $\sigma \in S_n$ is the product of disjoint cycles of lengths $n_1, n_2, \dots, n_r$ with $n_1 \leq n_2 \leq \dots \leq n_r$ (including its 1-cycles) then the integers $n_1, n_2, \dots, n_r$ are called the **cycle type** of σ .

(2) If $n \in \mathbb{Z}^+$, a partition of n is any nondecreasing sequence of positive integers whose sum is n .

Proposition 1.5. Two elements of S_n are conjugate in S_n if and only if they have the same cycle type. The number of conjugacy classes of S_n equals the number of partitions of n .

§2 The Alternating Group

Definition 2.1. Let $x_1, \dots, x_n$ be independent variables and let Δ be the polynomial

$$\Delta = \prod_{1 \leq i < j \leq n} (x_i - x_j)$$

For each $\sigma \in S_n$ let σ act on Δ by permuting the variables in the same way it permutes their indices:

$$\sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)})$$

or each $\sigma \in S_n$ let

$$\epsilon(\sigma) = \begin{cases} +1, & \text{if } \sigma(\Delta) = \Delta \\ -1, & \text{if } \sigma(\Delta) = -\Delta \end{cases}$$

Then

(1) $\epsilon(\sigma)$ is called the sign of σ .

(2) σ is called an **even permutation** if $\epsilon(\sigma) = 1$ and an **odd permutation** if $\epsilon(\sigma) = -1$

Proposition 2.2. (1) The map $\epsilon: S_n \rightarrow \{\pm 1\}$ is a homomorphism.

(2) Transpositions are all odd permutations and ϵ is a surjective homomorphism.

(3) An m -cycle is an odd permutation if and only if m is even.

(4) The permutation σ is odd if and only if the number of cycles of even length in its cycle decomposition is odd.

Definition 2.3. *The alternating group of degree n , denoted by A_n , is the kernel of the homomorphism ϵ (i.e., the set of even permutations).*

Chapter IV

Group Series

§1 Nilpotent Group

Definition 1.1. Let G be a group. The center $Z(G)$ of G is a normal subgroup. Let $Z_2(G)$ be the inverse image of $Z(G/Z(G))$ under the canonical projection $G \rightarrow G/Z(G)$. Then $Z_2(G)$ is normal in G and contains $Z(G)$. Continue this process by defining inductively: $Z_1(G) = Z(G)$ and $Z_i(G)$ is the inverse image of $Z(G/Z_{i-1}(G))$ under the canonical projection $G \rightarrow G/Z_{i-1}(G)$. Thus we obtain a sequence of normal subgroups of G , called the **ascending central series** of G :

$$\langle e \rangle < Z_1(G) < Z_2(G) < \cdots$$

A group G is **nilpotent** if $Z_n(G) = G$ for some n .

Theorem 1.2. Every finite p -group is nilpotent.

Proof. G and all its nontrivial quotients are p -groups, and therefore, have nontrivial centers by Corollary 5.4. This implies that if $G \neq C_i(G)$, then $C_i(G)$ is strictly contained in $C_{i+1}(G)$. Since G is finite, $C_n(G)$ must be G for some n . $\square$

Theorem 1.3. The direct product of a finite number of nilpotent groups is nilpotent.

Proof. Suppose for convenience that $G = H \times K$, the proof for more than two factors being similar. Assume inductively that $C_i(G) = C_i(H) \times C_i(K)$ (the case $i = 1$ is obvious). Let π_H be the canonical epimorphism $H \rightarrow H/C_i(H)$ and similarly for π_K . Verify that the canonical epimorphism $\varphi : G \rightarrow G/C_i(G)$ is the composition

$$G = H \times K \xrightarrow{\pi} H/C_i(H) \times K/C_i(K) \xrightarrow{\psi} \frac{H \times K}{C_i(H) \times C_i(K)} = \frac{H \times K}{C_i(H \times K)} = G/C_i(G),$$

where $\pi = \pi_H \times \pi_K$ (Theorem I.8.10), and ψ is the isomorphism of Corollary I.8.11. Consequently,

$$\begin{aligned} C_{i+1}(G) &= \varphi^{-1} [C(G/C_i(G))] = \pi^{-1} \psi^{-1} [C(G/C_i(G))] \\ &= \pi^{-1} [C(H/C_i(H) \times K/C_i(K))] \\ &= \pi^{-1} [C(H/C_i(H)) \times C(K/C_i(K))] \\ &= \pi_H^{-1} [C(H/C_i(H))] \times \pi_K^{-1} [C(K/C_i(K))] \\ &= C_{i+1}(H) \times C_{i+1}(K). \end{aligned}$$

Thus the inductive step is proved and $C_i(G) = C_i(H) \times C_i(K)$ for all i . Since H, K are nilpotent, there exists $n \in \mathbb{N}^*$ such that $C_n(H) = H$ and $C_n(K) = K$, whence $C_n(G) = H \times K = G$. Therefore, G is nilpotent. $\square$

Proposition 1.4. *If H is a proper subgroup of a nilpotent group G , then H is a proper subgroup of its normalizer $N_G(H)$.*

Proof. Let $C_0(G) = \langle e \rangle$ and let n be the largest index such that $C_n(G) < H$; (there is such an n since G is nilpotent and H a proper subgroup). Choose $a \in C_{n+1}(G)$ with $a \notin H$. Then for every $h \in H$,

$$C_n ah = (C_n a) (C_n h) = (C_n h) (C_n a) = C_n ha$$

in $G/C_n(G)$ since $C_n a$ is in the center of $C_{n+1}(G)$. Thus $ah = h'ha$, where $h' \in C_n(G) < H$. Hence $aha^{-1} \in H$ and $a \in N_G(H)$. Since $a \notin H$, H is a proper subgroup of $N_G(H)$. $\square$

Theorem 1.5. *A finite group is nilpotent if and only if it is the direct product of its Sylow subgroups.*

Proof. If G is the direct product of its Sylow p -subgroups, then G is nilpotent by Theorems 7.2 and 7.3.

If G is nilpotent and P is a Sylow p -subgroup of G for some prime p , then either $P = G$ (and we are done) or P is a proper subgroup of G . In the latter case P is a proper subgroup of $N_G(P)$ by Lemma 7.4. Since $N_G(P)$ is its own normalizer by Theorem 5.11, we must have $N_G(P) = G$ by Lemma 7.4. Thus P is normal in G , and hence the unique Sylow p -subgroup of G by Theorem 5.9.

Let $|G| = p_1^{n_1} \cdots p_k^{n_k}$ (p_i distinct primes, $n_i > 0$) and let $P_1, P_2, \dots, P_k$ be the corresponding (proper normal) Sylow subgroups of G . Since $|P_i| = p_i^{n_i}$ for each i , $P_i \cap P_j = \langle e \rangle$ for $i \neq j$. By Theorem I.5.3 $xy = yx$ for every $x \in P_i, y \in P_j$ ($i \neq j$). It follows that for each i , $P_1 P_2 \cdots P_{i-1} P_{i+1} \cdots P_k$ is a subgroup in which every element has order dividing $p_1^{n_1} \cdots p_{i-1}^{n_{i-1}} p_{i+1}^{n_{i+1}} \cdots p_k^{n_k}$. Consequently, $P_i \cap (P_1 \cdots P_{i-1} P_{i+1} \cdots P_k) = \langle e \rangle$ and $P_1 P_2 \cdots P_k = P_1 \times \cdots \times P_k$. Since $|G| = p_1^{n_1} \cdots p_k^{n_k} = |P_1 \times \cdots \times P_k| = |P_1 \cdots P_k|$ we must have $G = P_1 P_2 \cdots P_k = P_1 \times \cdots \times P_k$. $\square$

§2 Solvable Group

Definition 2.1. *Let G be a group, let $x, y \in G$ and let A, B be nonempty subsets of G .*

1. Define $[x, y] := x^{-1}y^{-1}xy$, called the **commutator of x and y** .
2. Define $[A, B] := \langle [a, b] \mid a \in A, b \in B \rangle$, the group generated by commutators of elements from A and from B .
3. Define $G^{(1)} := [G, G]$ be the **commutator subgroup of G** .
4. The sequence $G^{(n)} := [G^{(n-1)}, G^{(n-1)}]$ (with $G^{(0)} = G$) is called the **derived series of G** .
5. G is said to be **solvable** if there is a $n > 0$ such that $G^{(n)} = \langle e \rangle$ (equivalently, $G^{(n-1)}$ is abelian).

Proposition 2.2. Let G be a group and $N \trianglelefteq G$.

1. $G^{(1)} \trianglelefteq G$
2. G/N is abelian if and only if $G^{(1)} \leq N$

We now have the following characterization of solvable groups.

Theorem 2.3. Let G be a group of order n . Then G is solvable if and only if for any decomposition $n = st$ with $(s, t) = 1$, G has a subgroup of order s .

Proof. If G is solvable, then $G^{(n)} = \langle e \rangle$ for some $n > 0$. Let $N = G^{(n-1)}$. Then $N \trianglelefteq G$ and G/N is abelian. Since $(s, t) = 1$, there are subgroups of G/N of order s and t . The inverse image of the subgroup of order s under the canonical projection $G \rightarrow G/N$ is a subgroup of G of order s . □

Theorem 2.4. If N and G/N are solvable, then G is solvable.

Proof. To see this let $\bar{G} = G/N$, let $1 = N_0 \trianglelefteq N_1 \trianglelefteq \dots \trianglelefteq N_n = N$ be a chain of subgroups of N such that N_{i+1}/N_i is abelian, $0 \leq i < n$ and let $\bar{1} = \bar{G}_0 \trianglelefteq \bar{G}_1 \trianglelefteq \dots \trianglelefteq \bar{G}_m = \bar{G}$ be a chain of subgroups of $\bar{G}$ such that $\bar{G}_{i+1}/\bar{G}_i$ is abelian, $0 \leq i < m$.

By the Lattice Isomorphism Theorem there are subgroups G_i of G with $N \leq G_i$ such that $G_i/N = \bar{G}_i$ and $G_i \trianglelefteq G_{i+1}$, $0 \leq i < m$. By the Third Isomorphism Theorem

$$G_{i+1}/G_i \cong (G_{i+1}/N) / (G_i/N) = \bar{G}_{i+1}/\bar{G}_i$$

is abelian. Thus

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \dots \trianglelefteq N_n = N = G_0 \trianglelefteq G_1 \trianglelefteq \dots \trianglelefteq G_m = G$$

is a chain of subgroups of G all of whose successive quotient groups are abelian. This proves G is solvable. □

§3 Normal and Subnormal Series

Definition 3.1. A *subnormal series* of a group G is a chain of subgroups

$$G = G_0 > G_1 > \cdots > G_n$$

such that G_{i+1} is normal in G_i for $0 \leq i < n$. The **factors** of the series are the quotient groups G_i/G_{i+1} . The **length** of the series is the number of strict inclusions. A subnormal series such that G_i is normal in G for all i is said to be **normal**.

Definition 3.2. Let $G = G_0 > G_1 > \cdots > G_n$ be a subnormal series. A one-step refinement of this series is any subnormal series of the form $G = G_0 > \cdots > G_i > N > G_{i+1} > \cdots > G_n$ or $G = G_0 > \cdots > G_n > N$, where N is a normal subgroup of G_i and (if $i < n$) G_{i+1} is normal in N .

A **refinement** of a subnormal series S is any subnormal series obtained from S by a finite sequence of one-step refinements. A refinement of S is said to be **proper** if its length is larger than the length of S .

Two subnormal series S and T of a group G are **equivalent** if there is a one-to-one correspondence between the nontrivial factors of S and the nontrivial factors of T such that corresponding factors are isomorphic groups.

Definition 3.3. A subnormal series $G = G_0 > G_1 > \cdots > G_n = \langle e \rangle$ is a **composition series** if each factor G_i/G_{i+1} is simple.

A subnormal series $G = G_0 > G_1 > \cdots > G_n = \langle e \rangle$ is a **solvable series** if each factor is abelian.

Theorem 3.4. (1) Every finite group G has a composition series.

(2) Every refinement of a solvable series is a solvable series.

(3) A subnormal series is a composition series if and only if it has no proper refinements.

Proof. (1) Let G_1 be a maximal normal subgroup of G ; then G/G_1 is simple by Corollary I.5.12. Let G_2 be a maximal normal subgroup of G_1 , and so on. Since G is finite, this process must end with $G_n = \langle e \rangle$. Thus $G > G_1 > \cdots > G_n = \langle e \rangle$ is a composition series.

(2) If G_i/G_{i+1} is abelian and $G_{i+1} \triangleleft H \triangleleft G_i$, then H/G_{i+1} is abelian since it is a subgroup of G_i/G_{i+1} and G_i/H is abelian since it is isomorphic to the quotient $(G_i/G_{i+1}) / (H/G_{i+1})$ by the Third Isomorphism Theorem I.5.10. The conclusion now follows immediately.

(iii) If $G_{i+1} \triangleleft_{\neq} G_i$ are groups, then H/G_{i+1} is a proper normal subgroup of G_i/G_{i+1} and every proper normal subgroup of G_i/G_{i+1} has this form by Corollary I.5.12. The conclusion now follows from the observation that a subnormal series $G = G_0 > G_1 > \cdots > G_n = \langle e \rangle$ has a proper refinement if and only if there is a subgroup H such that for some i , $G_{i+1} \triangleleft_{\neq} G_i$. $\square$

Theorem 3.5. (1) A group G is solvable if and only if it has a solvable series.

(2) A finite group G is solvable if and only if G has a composition series whose factors are $\mathbb{Z}_p$ of order prime number p .

Proof. (1) If G is solvable, then the derived series $G > G^{(1)} > G^{(2)} > \cdots > G^{(n)} = \langle e \rangle$ is a solvable series by Theorem 7.8.

If $G = G_0 > G_1 > \cdots > G_n = \langle e \rangle$ is a solvable series for G , then G/G_1 abelian implies that $G_1 > G^{(1)}$ by Theorem 7.8; G_1/G_2 abelian implies $G_2 > G_1' > G^{(2)}$. Continue by induction and conclude that $G_2 > G^{(i)}$ for all i ; in particular $\langle e \rangle = G_n > G^{(n)}$ and G is solvable.

(2) A composition series with cyclic factors is a solvable series. Conversely, assume $G = G_0 > G_1 > \cdots > G_n = \langle e \rangle$ is a solvable series for G . If $G_0 \neq G_1$, let H_1 be a maximal normal subgroup of $G = G_0$ which contains G_1 . If $H_1 \neq G_1$, let H_2 be a maximal normal subgroup of H_1 which contains G_1 , and so on. Since G is finite, this gives a series $G > H_1 > H_2 > \cdots > H_k > G_1$ with each subgroup a maximal normal subgroup of the preceding, whence each factor is simple. Doing this for each pair (G_i, G_{i+1}) gives a solvable refinement $G = N_0 > N_1 > \cdots > N_r = \langle e \rangle$ of the original series by Theorem 8.4 (ii). Each factor of this series is abelian and simple and hence cyclic of prime order (Exercise I.4.3). Therefore, $G > N_1 > \cdots > N_r = \langle e \rangle$ is a composition series. $\square$

Lemma 3.6 (Zassenhaus). *Let A_2, A_1, B_2, B_1 be subgroups of a group G such that A_2 is normal in A_1 and B_2 is normal in B_1 .*

- (1) $A_2 (A_1 \cap B_2)$ is a normal subgroup of $A_2 (A_1 \cap B_1)$;
- (2) $B_2 (A_2 \cap B_1)$ is a normal subgroup of $B_2 (A_1 \cap B_1)$;
- (3) $A_2 (A_1 \cap B_1) / A_2 (A_1 \cap B_2) \cong B_2 (A_1 \cap B_1) / B_2 (A_2 \cap B_1)$.

Theorem 3.7 (Schreier). *Any two subnormal [resp. normal] series of a group G have subnormal [resp. normal] refinements that are equivalent.*

Proof. Let $G = G_0 > G_1 > \cdots > G_n$ and $G = H_0 > H_1 > \cdots > H_m$ be subnormal [resp. normal] series. Let $G_{n+1} = \langle e \rangle = H_{m+1}$ and for each $0 \leq i \leq n$ consider the groups

$$\begin{aligned} G_i &= G_{i+1} (G_i \cap H_0) > G_{i+1} (G_i \cap H_1) > \cdots > G_{i+1} (G_i \cap H_j) > G_{i+1} (G_i \cap H_{j+1}) \\ &> \cdots > G_{i+1} (G_i \cap H_m) > G_{i+1} (G_i \cap H_{m+1}) = G_{i+1}. \end{aligned}$$

For each $0 \leq j \leq m$, the Zassenhaus Lemma (applied to G_{i+1}, G_i, H_{j+1} , and H_j) shows that $G_{i+1} (G_i \cap H_{j+1})$ is normal in $G_{i+1} (G_i \cap H_j)$. [If the original series were both normal, then each $G_{i+1} (G_i \cap H_j)$ is normal in G by Theorem I.5.3 (iii) and Exercises I.5.2 and I.5.13.] Inserting these groups between each G_i and G_{i+1} , and denoting $G_{i+1} (G_i \cap H_j)$ by $G(i, j)$ thus gives a subnormal [resp. normal] refinement of the series $G_0 > G_1 > \cdots > G_n$:

$$\begin{aligned} G &= G(0, 0) > G(0, 1) > \cdots > G(0, m) > G(1, 0) > G(1, 1) > \\ &G(1, 2) > \cdots > G(1, m) > G(2, 0) > \cdots > G(n-1, m) > G(n, 0) > \cdots > G(n, m), \end{aligned}$$

where $G(i, 0) = G_i$. Note that this refinement has $(n+1)(m+1)$ (not necessarily distinct) terms. A symmetric argument shows that there is a refinement of $G = H_0 > H_1 > \cdots > H_m$ (where

$H(i, j) = H_{j+1}(G_i \cap H_j)$ and $H(0, j) = H_j$:

$$G = H(0, 0) > H(1, 0) > \cdots > H(n, 0) > H(0, 1) > H(1, 1) > H(2, 1) > \cdots > \\ H(n, 1) > H(0, 2) > \cdots > H(n, m-1) > H(0, m) > \cdots > H(n, m).$$

This refinement also has $(n+1)(m+1)$ terms. For each pair $(i, j)(0 \leq i \leq n, 0 \leq j \leq m)$ there is by the Zassenhaus Lemma 8.9 (applied to G_{i+1}, G_i, H_{j+1} , and H_j) an isomorphism:

$$\frac{G(i, j)}{G(i, j+1)} = \frac{G_{i+1}(G_i \cap H_j)}{G_{i+1}(G_i \cap H_{j+1})} \cong \frac{H_{j+1}(G_i \cap H_j)}{H_{j+1}(G_{i+1} \cap H_j)} = \frac{H(i, j)}{H(i+1, j)}.$$

This provides the desired one-to-one correspondence of the factors and shows that the refinements are equivalent. $\square$

Theorem 3.8 (Jordan-Hölder). *Any two composition series of a group G are equivalent. Therefore every group having a composition series determines a unique list of simple groups.*

Chapter V

Structure of Groups

§1 Free Groups

§1.1 Words on Free Group

Theorem 1.1. *Let G be a group, X a set and $\varphi : S \rightarrow G$ a map. Then there is a unique group homomorphism $\Phi : F(S) \rightarrow G$ such that the following diagram commutes:*

$$\begin{array}{ccc} S & \xrightarrow{i} & F(S) \\ & \searrow \phi & \downarrow \Phi \\ & & G \end{array}$$

Corollary 1.2. *Every group G is a homomorphic image of a free group.*

Corollary 1.3. *$F(S)$ is unique up to a unique isomorphism which is the identity map on the set S .*

Definition 1.4. *The group $F(S)$ is called the **free group on the set S** . A group F is a free group if there is some set S such that $F = F(S)$ - in this case we call S a set of **free generators (or a free basis)** of F . The cardinality of S is called the **rank of the free group**.*

Theorem 1.5 (Schreier). *Subgroups of a free group are free.*

§1.2 Presentations

Lemma 1.6. *Let G be a group and a subset S of G , the normal closure of S (**normal subgroup generated by S**) is defined by the*

$$\bigcap_{S \subset N \trianglelefteq G} N$$

Definition 1.7. *Let X be a set and Y a set of (reduced) words on X . A group G is said to be the group determined by the **generators** $x \in X$ and **relations** $w = e$ ($w \in Y$) provided $G \cong G/N$,*

where F is the free group on X and N the normal subgroup of F generated by Y . One says that $(X \mid Y)$ is a **presentation** of G .

We say G is **finitely generated** if there is a presentation (S, R) such that S is finite. And we say G is **finitely presented** if there is a presentation (S, R) with both S and R finite.

If G is finitely presented with $S = \{s_1, \dots, s_n\}$ and $R = \{w_1, \dots, w_k\}$, we write:

$$G = \langle s_1, s_2, \dots, s_n \mid w_1 = w_2 = \dots = w_k = 1 \rangle$$

Theorem 1.8 (Van Dyck). *Let X be a set, Y a set of (reduced) words on X and G the group defined by the generators $x \in X$ and relations $w = e(w \in Y)$. If H is any group such that $H = \langle X \rangle$ and H satisfies all the relations $w = e(w \in Y)$, then there is an epimorphism $G \rightarrow H$.*

Proof. If F is the free group on X then the inclusion map $X \rightarrow H$ induces an epimorphism $\varphi : F \rightarrow H$ by Corollary 9.3. Since H satisfies the relations $w = e(w \in Y)$, $Y \subset \text{Ker } \varphi$. Consequently, the normal subgroup N generated by Y in F is contained in $\text{Ker } \varphi$. By Corollary 5.8 φ induces an epimorphism $F/N \rightarrow H/0$. Therefore the composition $G \cong F/N \rightarrow H/0 \cong H$ is an epimorphism. $\square$

Theorem 1.9. *Every finite group is finitely presented.*

Proof: To see this let $G = \{g_1, \dots, g_n\}$ be a finite group. Let $S = G$ and let $\pi : F(S) \rightarrow G$ be the homomorphism extending the identity map of S . Let

$$R_0 = \{g_i g_j g_k^{-1} : i, j, k = 1, \dots, n \text{ and } g_i g_j = g_k\}$$

Clearly $R_0 \leq \text{ker } \pi$.

If N is the normal closure of R_0 in $F(S)$ and $\tilde{G} = F(S)/N$, then $N \leq \text{ker } \pi$ and G is a homomorphic image of $\tilde{G}$ (i.e., π factors through N). Moreover, the set of elements $\{\tilde{g}_i \mid i = 1, \dots, n\}$ is closed under multiplication. Since this set generates $\tilde{G}$, it must equal $\tilde{G}$. Thus $|\tilde{G}| = |G|$ and so $N = \text{ker } \pi$ and (S, R_0) is a presentation of G .

This illustrates a sufficient condition for (S, R) to be a presentation for a given finite group G : (i) S must be a generating set for G , and (ii) any group generated by S satisfying the relations in R must have order $\leq |G|$.

§2 Product and Coproduct

§2.1 Direct Product

Definition 2.1. *Let $\{G_i \mid i \in I\}$ be an arbitrary family of groups. Define a binary operation on the Cartesian product*

$$\prod_{i \in I} G_i = \{(g_i)_{i \in I} : g(i) \in G_i\}$$

by

$$(g_i)_{i \in I} \cdot (h_i)_{i \in I} = (g_i \cdot h_i)_{i \in I}$$

is called the **direct product** of the family of groups $\{G_i \mid i \in I\}$.

Remark. It equivalent that

$$\prod_{i \in I} G_i \cong \left\langle \bigsqcup_{i \in I} G_i \mid \bigsqcup_{i \in I} R_i, g_i g_j \right\rangle$$

isomorphism.

Proposition 2.2. If $\{G_i \mid i \in I\}$ is a family of groups, then

1. The direct product $\prod_{i \in I} G_i$ is a group;
2. For each $k \in I$, the map $\pi_k : \prod_{i \in I} G_i \rightarrow G_k$ given by $f \mapsto f(k)$ is an epimorphism of groups. The maps π_k are called the **canonical projections** of the direct product.

Theorem 2.3. Let $\{G_i : i \in I\}$ be a family of groups and $\{f_i : G \rightarrow G_i \mid i \in I\}$ a family of group homomorphisms. Then there is a unique homomorphism $f : G \rightarrow \prod_{i \in I} G_i$ such that the following diagram

$$\begin{array}{ccc} G & & \\ \downarrow \exists! f & \searrow f_j & \\ \prod_{i \in I} G_i & \xrightarrow{\pi_j} & G_j \end{array}$$

is commutative for all $i \in I$ and this property determines $\prod_{i \in I} G_i$ uniquely up to isomorphism. In other words, $\prod_{i \in I} G_i$ is a product in the category of groups.

§2.2 Free Product

Definition 2.4. Let $\{G_i \mid i \in I\}$ be an arbitrary family of groups with presentation $\langle S_i \mid R_i \rangle$. We define

$$*_{i \in I} G_i = \left\langle \bigsqcup_{i \in I} S_i \mid \bigsqcup_{i \in I} R_i \right\rangle$$

§2.3 Weak Direct Product

Definition 2.5. Let $\{G_i \mid i \in I\}$ be an arbitrary family of groups. The **weak direct product** of a family of groups $\{G_i \mid i \in I\}$, denoted

$$\prod_{i \in I}^w G_i = \left\{ f \in \prod_{i \in I} G_i : f(i) = e_i \text{ for all but a finite number of } i \in I \right\}$$

If all the groups G_i are (additive) abelian, $\prod_{i \in I}^w G_i$ is usually called the **direct sum** and is denoted $\sum_{i \in I} G_i$.

Theorem 2.6. If $\{G_i \mid i \in I\}$ is a family of groups, then

- (1) $\prod_{i \in I}^w G_i$ is a normal subgroup of $\prod_{i \in I} G_i$
 (2) for each $k \in I$, the map

$$\iota_k : G_k \rightarrow \prod_{i \in I}^w G_i$$

given by $\iota_k(a) = \{a_i\}_{i \in I}$, where $a_i = e$ for $i \neq k$ and $a_k = a$, is a monomorphism of groups;

- (3) for each $i \in I$, $\iota_i(G_i)$ is a normal subgroup of $\prod_{i \in I} G_i$.

The maps ι_i are called the **canonical injections**

Theorem 2.7. Let $\{A_i : i \in I\}$ be a family of abelian groups (written additively). If A is an abelian group and $\{f_i : A_i \rightarrow A : i \in I\}$ a family of homomorphisms, then there is a unique homomorphism $f : \sum_{i \in I} A_i \rightarrow A$ such that

$$\begin{array}{ccc} & A & \\ f \uparrow & \nwarrow f_j & \\ \sum_{i \in I} A_i & \xleftarrow{\iota_j} & G_j \end{array}$$

for all $j \in I$ and this property determines $\sum_{i \in I} A_i$ uniquely up to isomorphism. In other words, $\sum_{i \in I} A_i$ is a coproduct in the category of abelian groups.

Theorem 2.8. Let $\{N_i \mid i \in I\}$ be a family of normal subgroups of a group G such that

- (i) $G = \langle \bigcup_{i \in I} N_i \rangle$;
 (ii) for each $k \in I$, $N_k \cap \langle \bigcup_{i \neq k} N_i \rangle = \langle e \rangle$.

Then $G \cong \prod_{i \in I}^w N_i$, G is said to be the **internal weak direct product** of the family $\{N_i \mid i \in I\}$. (or the **internal direct sum** if G is abelian)

Corollary 2.9. If $N_1, N_2, \dots, N_r$ are normal subgroups of a group G such that $G = N_1 N_2 \cdots N_r$ and for each $1 \leq k \leq r$, $N_k \cap (N_1 \cdots N_{k-1} N_{k+1} \cdots N_r) = \langle e \rangle$, then $G \cong N_1 \times N_2 \times \cdots \times N_r$.

Theorem 2.10. Let $\{N_i \mid i \in I\}$ be a family of normal subgroups of a group G . Then G is the internal weak direct product of the family $\{N_i \mid i \in I\}$ if and only if every nonidentity element of G is a unique product

$$n_{i_1} n_{i_2} \cdots n_{i_k}$$

with $i_1, \dots, i_k$ distinct indexes of I and $e \neq n_{i_s} \in N_{i_s}$ for each $s = 1, 2, \dots, k$.

§3 The Krull-Schmidt Theorem

Definition 3.1. A group G is **indecomposable** if $G \neq \langle e \rangle$ and G is not the (internal) direct product of two of its proper subgroups.

Proposition 3.2. Let G be group.

1. G is indecomposable if and only if $G \neq \langle e \rangle$ and $G \cong H \times K$ implies $H = \langle e \rangle$ or $K = \langle e \rangle$.
2. Every simple group is indecomposable.
3. However indecomposable groups need not be simple: $\mathbb{Z}, \mathbb{Z}_{p^n}$ (p prime) and S_n are indecomposable but not simple.

Definition 3.3. A group G is said to satisfy the **ascending chain condition (ACC) on subgroups [resp. normal subgroup]** if for every chain $G_1 < G_2 < \dots$ of subgroups [resp. normal subgroups] of G there is an integer n such that $G_i = G_n$ for all $i \geq n$.

G is said to satisfy the **descending chain condition (DCC) on subgroups [resp. normal subgroup]** if for every chain $G_1 > G_2 > \dots$ of subgroups [resp. normal subgroup] of G there is an integer n such that $G_i = G_n$ for all $i \geq n$.

Theorem 3.4 (Krull-Schmidt Theorem). *If a group G satisfies either the ascending or descending chain condition on normal subgroups, then G is the direct product of a finite number of indecomposable subgroups.*

§4 The Fundamental Theorem of Finitely Generated Abelian Groups

Part II

Ring Theory

Chapter VI

Ring Theory

§1 Basic Definition

Definition 1.1. A **ring** A is a set together with two binary operations $+$ and $\times$ satisfying the following axioms:

- (i) $(A, +)$ is an abelian group,
- (ii) $\times$ is associative
- (iii) the distributive laws hold in A : for all $a, b, c \in A$

$$(a + b) \times c = (a \times c) + (b \times c) \quad \text{and} \quad a \times (b + c) = (a \times b) + (a \times c)$$

The ring A is **commutative** if

- (iv) multiplication $\times$ is commutative.

The ring A is called **unital** if it has an **identity** 1_A s.t.

- (v) $1_A \times a = a \times 1_A = a$ for all $a \in A$.

Remark. In this book, we usually use the term "ring" to refer a unital ring.

Definition 1.2. A **subring** of A is a subgroup of A that is closed under multiplication and contains the identity element 1_A .

Proposition 1.3. Let A be a ring. Then

1. $0a = a0 = 0$ for all $a \in R$.
2. $(-a)b = a(-b) = -(ab)$ for all $a, b \in R$.
3. $(-a)(-b) = ab$ for all $a, b \in R$.
4. the identity is unique and $-a = (-1)a$.

Definition 1.4. Let A be a ring.

1. A nonzero element a of A is called a **zero divisor** if there is a nonzero element b in A such that either $ab = 0$ or $ba = 0$.
2. An element u of A is called a **unit** in A if there is some v in A such that $uv = vu = 1$. The set of units in A is denoted $A^\times$.

Definition 1.5. A commutative ring with identity $1 \neq 0$ is called an **integral domain** if it has no zero divisors.

Proposition 1.6 (Cancellation property). Let R be a ring. Then

1. Assume a, b and c are elements of R with a not a zero divisor. If $ab = ac$, then either $a = 0$ or $b = c$.
2. In particular, if a, b, c are any elements in an integral domain and $ab = ac$, then either $a = 0$ or $b = c$.
3. Any finite integral domain is a field.

Definition 1.7. Let A be a ring. If there is a least positive integer n such that $nr = 0$ for all $r \in A$, then A is said to have **characteristic** n . If no such n exists A is said to have characteristic zero. (Notation: $\text{char } A = n$).

Theorem 1.8. Let A be a ring with identity 1_A and characteristic $n > 0$.

1. If $\varphi : \mathbb{Z} \rightarrow A$ is the map given by $m \mapsto m1_A$, then φ is a homomorphism of rings with kernel $\langle n \rangle = n\mathbb{Z}$.
2. n is the least positive integer such that $n1_A = 0$.
3. If A has no zero divisors (in particular if A is an integral domain), then n is prime.

§2 Ideal

§2.1 Definition and Quotient Ring

Definition 2.1. Let A be a ring. A subset $\mathfrak{a}$ of A is a **left ideal** of A if

- (i) $\mathfrak{a}$ is an additive subgroup of A ,
- (ii) $\mathfrak{a}$ is closed under left multiplication by elements from A , i.e., $r\mathfrak{a} \subseteq \mathfrak{a}$ for all $r \in A$.

A subset $\mathfrak{b}$ that is both a left ideal and a right ideal is called an **ideal** (or a **two-sided ideal**) of A . An ideal $\mathfrak{b}$ is **proper** if $\mathfrak{b} \neq A$ and $\mathfrak{b} \neq 0$. The ideal $\{0\}$ is called the **trivial ideal** and is denoted by 0 .

Theorem 2.2. *Let A be a ring. A nonempty subset $\mathfrak{a}$ of a ring A is a left [resp. right] ideal if and only if for all $a, b \in \mathfrak{a}$ and $r \in A$:*

$$(i) \quad a, b \in \mathfrak{a} \quad \Rightarrow \quad a - b \in \mathfrak{a}$$

$$(ii) \quad a \in \mathfrak{a}, r \in A \quad \Rightarrow \quad ra \in \mathfrak{a} \text{ [resp. } ar \in \mathfrak{a}]$$

Corollary 2.3. *Let R be a ring.*

1. *If $\{I_\alpha\}$ is a family of ideals [resp. left ideal] in a ring R , then*

$$\bigcap I_\alpha$$

is also a ideal [resp. left ideal].

2. *If $\{J_\beta\}$ is a chain of ideals [resp. left ideal] in $\mathcal{P}(R)$ i.e. either $J_\beta \subset$ or $J_{\beta'} \subset J_\beta$ holds for all β, β' , then*

$$\bigcup J_\beta$$

is also a ideal [resp. left ideal] of R .

Definition 2.4. *Let A be a ring and let $\mathfrak{a}$ be an ideal of A . Then the (additive) quotient group $A/\mathfrak{a}$ is a ring called the **quotient ring of A by $\mathfrak{a}$** under the binary operations:*

$$(r + \mathfrak{a}) + (s + \mathfrak{a}) = (r + s) + \mathfrak{a} \quad \text{and} \quad (r + \mathfrak{a}) \times (s + \mathfrak{a}) = (rs) + \mathfrak{a}$$

for all $r, s \in A$. Conversely, if $\mathfrak{b}$ is any subgroup such that the above operations are well defined, then $\mathfrak{b}$ is an ideal of A .

Remark. *If $\mathfrak{a}$ is a left [resp. right] ideal of A , then the quotient group $A/\mathfrak{a}$ is left [resp. right] A -module under the operation*

§2.2 Ideals generated by a set

Definition 2.5. *Let $X_1, X_2, \dots, X_n$ be nonempty subsets of a ring A . Then*

1. *Denote by $X_1 + X_2 + \dots + X_n$ the set*

$$\{x_1 + x_2 + \dots + x_n : x_i \in X_i, i = 1, 2, \dots, n\}$$

2. *If X and Y are nonempty subsets of A let XY denote the set of all finite sums*

$$\{x_1 y_1 + \dots + x_n y_n : n \in \mathbb{Z}_{\geq 1}; x_i \in X, y_i \in Y\}$$

If X consists of a single element x , we write xY for XY . Similarly if $Y = \{y\}$, we write Xy for XY .

Theorem 2.6. Let $A, A_1, A_2, \dots, A_n, B$ and C be subsets of ring A . Then

1. $(A + B) + C = A + (B + C); A + B = B + A$
2. $(AB)C = A(BC)$
3. $B(A_1 + A_2) = BA_1 + BA_2$; and $(A_1 + A_2)C = A_1C + A_2C$.

Definition 2.7. Let X be a subset of a ring R . Let $\{I_\alpha\}$ be the family of all (left) ideals in R which contain X . Then $\bigcap I_\alpha$ is called the **(left) ideal generated by X** . This ideal is denoted (X) . The elements of X are called **generators** of the ideal (X) .

If $X = \{x_1, \dots, x_n\}$, then the ideal (X) is denoted by $(x_1, x_2, \dots, x_n)$ and said to be **finitely generated**.

An ideal (x) generated by a single element is called a **principal ideal**. A principal ideal ring is a ring in which every ideal is principal. A principal ideal ring which is an integral domain is called a **principal ideal domain**.

Proposition 2.8. Let A be a ring and $X \subset A$, then

1. the left [resp. right] ideal generated by X is

$$(X)_l = AX \quad [\text{resp.}] \quad (X)_r = XA$$

the (two-sided) ideal generated by X is

$$(X) = AX + XA + AXA$$

2. if A is commutative, $(X)_l = (X)_r = (X) = AX$

§2.3 Prime ideal

Definition 2.9. Let A be a ring. An ideal $\mathfrak{p}$ in A is said to be **prime** if

- (i) $\mathfrak{p} \neq A$
- (ii) for any ideals A, B in A , $\mathfrak{a}\mathfrak{b} \subset \mathfrak{p} \Rightarrow \mathfrak{a} \subset \mathfrak{p}$ or $\mathfrak{b} \subset \mathfrak{p}$

The set of all prime ideals in a ring A is called the **spectrum** of A , denoted by $\text{Spec}(A)$.

Theorem 2.10. Let A be a ring and an ideal $\mathfrak{p} \neq A$.

1. If $ab \in \mathfrak{p} \Rightarrow a \in \mathfrak{p}$ or $b \in \mathfrak{p}$ for all $a, b \in A$, then $\mathfrak{p}$ is prime.

If A is commutative,

2. ideal $\mathfrak{p}$ is prime if and only if then $\mathfrak{p}$ satisfies the above condition.

Corollary 2.11. *Let A be a commutative ring and $\mathfrak{p}$ be an ideal in A . The following conditions are equivalent.*

1. *Ideal $\mathfrak{p}$ is prime*
2. *$A - \mathfrak{p}$ is a multiplicative set of A .*
3. *$A/\mathfrak{p}$ is an integral domain.*

Proposition 2.12. *Let K be a subring of a commutative ring R . If $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ are prime ideals of R such that $K \subset \mathfrak{p}_1 \cup \mathfrak{p}_2 \cup \dots \cup \mathfrak{p}_n$, then $K \subset \mathfrak{p}_i$ for some i .*

Proof. Assume $K \not\subset \mathfrak{p}_i$ for every i . It then suffices to assume that $n > 1$ and n is minimal; that is, for each i , $K \not\subset \bigcup_{j \neq i} \mathfrak{p}_j$. For each i there exists $a_i \in K - \bigcup_{j \neq i} \mathfrak{p}_j$. Since $K \subset \bigcup_i \mathfrak{p}_i$, each $a_i \in \mathfrak{p}_i$. The element $a_1 + a_2 a_3 \cdots a_n$ lies in K and hence in $\bigcup_i \mathfrak{p}_i$. Therefore $a_1 + a_2 a_3 \cdots a_n = b_j$ with $b_j \in \mathfrak{p}_j$. If $j > 1$, then $a_1 \in \mathfrak{p}_j$, which is a contradiction. If $j = 1$, then $a_2 a_3 \cdots a_n \in \mathfrak{p}_1$, whence $a_i \in \mathfrak{p}_1$ for some $i > 1$ by 2.11. $\square$

§2.4 Maximal ideal

Definition 2.13. *Let A be a ring. An ideal $\mathfrak{m}$ is called a **maximal ideal** [resp. maximal left ideal] if*

- (i) $\mathfrak{m} \neq A$
- (ii) *the only ideals [resp. left ideal] containing $\mathfrak{m}$ are A and $\mathfrak{m}$.*

Theorem 2.14. *In a ring A with identity $1_A \neq 0$, every ideal [resp. left ideal] in A (except A itself) is contained in a [resp. left ideal] maximal ideal.*

Proof. It follows from Zorn's lemma and corollary 2.3 $\square$

Theorem 2.15. *If A is a commutative ring, then every maximal ideal $\mathfrak{m}$ is prime.*

Theorem 2.16. *Let $\mathfrak{m}$ be an ideal in a ring A with identity $1_A \neq 0$.*

1. *If $\mathfrak{m}$ is maximal and A is commutative, then the quotient ring $A/\mathfrak{m}$ is a field.*
2. *If the quotient ring $A/\mathfrak{m}$ is a division ring, then $\mathfrak{m}$ is maximal.*

§2.5 Chinese Remainder Theorem

Assume that A is a commutative ring with identity.

Definition 2.17. *The ideals $\mathfrak{a}$ and $\mathfrak{b}$ of the ring A are said to be **comaximal** if $\mathfrak{a} + \mathfrak{b} = A$.*

Lemma 2.18. *If $\mathfrak{a}_1$ and $\mathfrak{a}_2$ are comaximal ideals in A , then $\mathfrak{a}_1 \mathfrak{a}_2 = \mathfrak{a}_1 \cap \mathfrak{a}_2$.*

Theorem 2.19 (Chinese Remainder Theorem). *Let A be a commutative ring with $1_A \neq 0$ and $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_k$ be ideals in A . Then*

1. *The map*

$$A \rightarrow A/\mathfrak{a}_1 \times A/\mathfrak{a}_2 \times \cdots \times A/\mathfrak{a}_k$$

defined by

$$r \mapsto (r + \mathfrak{a}_1, r + \mathfrak{a}_2, \dots, r + \mathfrak{a}_k)$$

is a ring homomorphism with kernel $\mathfrak{a}_1 \cap \mathfrak{a}_2 \cap \cdots \cap \mathfrak{a}_k$.

2. *If $\mathfrak{a}_i$ are comaximal in pairs, then this map is surjective and*

$$A/(\mathfrak{a}_1 \mathfrak{a}_2 \cdots \mathfrak{a}_k) = A/(\mathfrak{a}_1 \cap \mathfrak{a}_2 \cap \cdots \cap \mathfrak{a}_k) \cong A/\mathfrak{a}_1 \times A/\mathfrak{a}_2 \times \cdots \times A/\mathfrak{a}_k$$

Corollary 2.20. *Let n be a positive integer and let $p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ be its factorization into powers of distinct primes. Then*

$$\mathbb{Z}/n\mathbb{Z} \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})$$

as rings, so in particular we have the following isomorphism of multiplicative groups:

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})^\times.$$

§3 Homomorphisms

Definition 3.1. *Let R and S be ring.*

1. *A **ring homomorphism** is a map $f : R \rightarrow S$ satisfying*

$$f(a + b) = f(a) + f(b), \quad f(ab) = f(a)f(b)$$

for all $a, b \in R$ and $f(1_R) = 1_S$.

2. *The **kernel** of the ring homomorphism f , denoted $\text{Ker } f$, is the set of elements of R that map to 0 in S .*

3. *A bijective ring homomorphism is called an **isomorphism**.*

Theorem 3.2 (The First Isomorphism Theorem for Rings). *If $f : R \rightarrow S$ is a homomorphism of rings, then*

1. *The kernel of f is an ideal of R .*

2. *The image of f is a subring of S and $R/\text{Ker } f \cong f(R)$.*

Corollary 3.3. *If I is any ideal of R , then the **natural projection***

$$R \rightarrow R/I \quad \text{defined by} \quad r \mapsto r + I$$

is a surjective ring homomorphism with kernel I . Thus every ideal is the kernel of a ring homomorphism.

Theorem 3.4 (The Second Isomorphism Theorem for Rings). *Let R be a ring. Let A be a subring and let B be an ideal of R . Then $A + B = \{a + b \mid a \in A, b \in B\}$ is a subring of R , $A \cap B$ is an ideal of A and*

$$(A + B)/B \cong A/(A \cap B)$$

Theorem 3.5 (The Third Isomorphism Theorem for Rings). *Let I and J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and*

$$(R/I)/(J/I) \cong R/J$$

Theorem 3.6 (The Fourth or Lattice Isomorphism Theorem for Rings). *Let $\mathfrak{a}$ be an ideal of ring R .*

1. *The correspondence*

$$R \leftrightarrow R/\mathfrak{a} \tag{1}$$

is an inclusion preserving bijection between the collection of subrings of R that contain I and the collection of subrings of $R/\mathfrak{a}$.

2. *Furthermore, a subring A containing $\mathfrak{a}$ is an ideal of R if and only if $A/\mathfrak{a}$ is an ideal of $R/\mathfrak{a}$.*

§4 Rings of Polynomial and Formal Power Series

We only consider $A[x]$ where A is a commutative ring with identity.

Proposition 4.1. *Let A be a ring with identity and $f = \sum_{i=0}^{\infty} a_i x^i \in A[[x]]$.*

1. *f is a unit in $A[[x]]$ if and only if its constant term a_0 is a unit in A*
2. *If a_0 is irreducible in A , then f is irreducible in $A[[x]]$.*

Corollary 4.2. *If A is a field, then $A[[x]]$ is a local ring with unique maximal ideal (x)]*

Chapter VII

Factorization in Integral Domains

§1 Divisor Decomposition

§1.1 Basic definition

Definition 1.1. Let R be an integral domain and let $a, b \in R$ with $b \neq 0$.

1. a is said to be a **multiple** of b if there exists an element $x \in R$ with $a = bx$. In this case b is said to divide a or be a **divisor** of a , written $b \mid a$.
2. If $a \mid b$ and $b \mid a$, then a and b are said to be **associates**.
3. If $b \mid a$ and $a \nmid b$ then b is called a **proper divisor** of a . Every element a has two **trivial divisors**: units and associate elements of a .

Theorem 1.2. Let R be an integral domain, and a, b, u be elements of R .

1. $a \mid b$ if and only if $(b) \subset (a)$.
2. a and b are associates $\Leftrightarrow (a) = (b) \Leftrightarrow a = bu$ for some unit u .

Definition 1.3. Let R be a commutative ring with $1_R \neq 0$.

1. Suppose $r \in R$, then r is called **irreducible** in R if

- (i) $r \neq 0$ and $r \notin R^\times$
- (ii) $r = ab \Rightarrow a$ or b is a unit.

The only divisors of an irreducible element are its associates and the units of R .

2. The element $p \in R$ is called **prime** in R if

- (i) $p \neq 0$ and $p \notin R^\times$
- (ii) if $p \mid ab$ for any $a, b \in R$, then $p \mid a$ or $p \mid b$.

Remark. Every associate of an irreducible [resp. prime] element is irreducible [resp. prime].

Theorem 1.4. Let p and c be nonzero elements in an integral domain R .

1. p is prime if and only if (p) is nonzero prime ideal
2. c is irreducible if and only if (c) is maximal in the set S of all proper principal ideals of R .
3. Every prime element of R is irreducible.

Definition 1.5. Let X be a nonempty subset of a integral domain R

1. An element $d \in R$ is a **greatest common divisor** of X provided

- (i) $d \mid x$ for all $x \in X$
- (ii) if $d' \mid x$ for all $x \in X$ then $d' \mid d$.

A greatest common divisor of a and b will be denoted by $\gcd(a, b)$, or (a, b) .

2. An element $l \in R$ is a **least common multiple** of X such that

- (i) $x \mid l$ for all $x \in X$
- (ii) if $x \mid l'$ for all $x \in X$, then $l \mid l'$

A least common multiple of a and b will be denoted by $\text{lcm}(a, b)$, or $[a, b]$.

Remark. The greatest common divisor and least common multiple are unique up to association.

Definition 1.6. Let R be a integral domain, and $a_1, a_2, \dots, a_n$ have 1_R as a greatest common divisor, then $a_1, a_2, \dots, a_n$ are said to be **relatively prime**.

Theorem 1.7. Let $a_1, \dots, a_n$ be elements of a commutative ring R with identity. Then $d \in R$ is a greatest common divisor of $\{a_1, \dots, a_n\}$ and $d \in (a_1) + (a_2) + \dots + (a_n)$ if and only if $(d) = (a_1) + (a_2) + \dots + (a_n)$;

If F is a field, then x and y are relatively prime in the polynomial domain $F[x, y]$, but $F[x, y] = (1_F) \supsetneq (x) + (y)$

§2 E.D $\Rightarrow$ P.I.D $\Rightarrow$ U.F.D

§2.1 Unique Factorization Domain

Definition 2.1. A **unique factorization domain** is an integral domain R in which every nonzero element $r \in R$ which is not a unit has the following two properties:

- (i) every nonzero nonunit element a of R can be written $a = c_1 c_2 \cdots c_n$, with $c_1, \dots, c_n$ irreducible.

- (ii) If $a = c_1 c_2 \cdots c_n$ and $a = d_1 d_2 \cdots d_n$ (c_i, d_i irreducible), then $n = m$ and for some permutation $\sigma \in \mathfrak{S}_n$, c_i and $d_{\sigma(i)}$ are associates for every i .

Proposition 2.2. *If R is a unique factorization domain then*

1. *Irreducible and prime elements coincide.*
2. *for any $a_1, a_2, \dots, a_n$, there exists an unique greatest common divisor of $a_1, \dots, a_n$ in the sense of association and $(d) = (a_1) + (a_2) + \cdots + (a_n)$,*

§2.2 Principal rings and principal domains

Definition 2.3. *A **principal ideal ring** is a ring in which every ideal is principal.*

Theorem 2.4. *If R is principal ideal domain*

1. *R is a unique factorization domain thus proposition 2.2 holds.*
2. *maximal ideals and prime ideals coincide.*
3. *if $(a) + (b) = (c)$, then c is a greatest common divisor of a and b .*

Definition 2.5. *Define N to be a **Dedekind-Hasse norm** in commutative ring R if N is a positive norm and for every nonzero $a, b \in R$ either a is an element of the ideal (b) or there is a nonzero element in the ideal $(a, b) = (a) + (b)$ of norm strictly smaller than the norm of b . (i.e., either b divides a in R or there exist $s, t \in R$ with $0 < N(sa - tb) < N(b)$).*

Theorem 2.6. *The integral domain R is a P.I.D. if and only if R has a Dedekind-Hasse norm.*

Proof. Suppose that R has a Dedekind-Hasse norm N . Let $\mathfrak{a}$ be a nonzero ideal in R and let b be a nonzero element of $\mathfrak{a}$ of minimum norm. If $a \in \mathfrak{a}$, then either $a \in (b)$ or there exist $s, t \in R$ with $0 < N(sa - tb) < N(b)$. The latter is impossible since $sa - tb \in \mathfrak{a}$ and b has minimum norm in $\mathfrak{a}$. Therefore, $a \in (b)$ and consequently, $\mathfrak{a} = (b)$. Thus R is a principal ideal domain.

The converse is obvious. □

§2.3 Euclidean Ring and Euclidean domain

Definition 2.7. *Let R a commutative ring. R is a **Euclidean ring** if there is a function $\varphi : R - \{0\} \rightarrow \mathbb{N}$ such that:*

- (i) *if $a, b \in R$ and $ab \neq 0$, then $\varphi(a) \leq \varphi(ab)$;*
- (ii) *if $a, b \in R$ and $b \neq 0$, then there exist $q, r \in R$ such that $a = qb + r$ with $r = 0$, or $r \neq 0$ and $\varphi(r) < \varphi(b)$.*

*A Euclidean ring which is an integral domain is called a **Euclidean domain**.*

Theorem 2.8. *If R is a Euclidean ring, then*

1. *R is a principal ideal ring with identity.*
2. *if $\mathfrak{a}$ is any nonzero ideal in the Euclidean ring R with φ and $\mathfrak{a} = (a)$, then a is a nonzero element of $\mathfrak{a}$ of minimum norm.*

Proof. If I is a nonzero ideal in R , choose $a \in I$ such that $\varphi(a)$ is the least integer in the set of nonnegative integers $\{\varphi(x) \mid x \neq 0; x \in I\}$. If $b \in I$, then $b = qa$. Consequently, $I \subset Ra \subset (a) \subset I$. Therefore $I = Ra = (a)$ and R is a principal ideal ring.

Conversely, if $\mathfrak{a} = (a)$ and b is a nonzero element of $\mathfrak{a}$ of minimum norm, then $a = xy$ and $b = ya$ for some $x, y \in R$, whence $\varphi(a) = \varphi(xb) \geq \varphi(b)$ and $\varphi(b) = \varphi(ya) \geq \varphi(a)$. Thus a is also a nonzero element of $\mathfrak{a}$ of minimum norm.

Since R itself is an ideal, $R = Ra$ for some $a \in R$. Consequently, $a = ea = ae$ for some $e \in R$. If $b \in R = Ra$, then $b = xa$ for some $x \in R$. Therefore, $be = (xa)e = x(ae) = xa = b$, whence e is a multiplicative identity element for R . $\square$

Corollary 2.9. *Let R be a Euclidean ring and $a \in R$. Then $\varphi(1_R)$ is minimum and element a is a unit in R if and only if $\varphi(a) = \varphi(1_R)$.*

§3 Factorization in Polynomial Rings

Definition 3.1.

Theorem 3.2. *Let R be a ring and $f, g \in R[x_1, \dots, x_n]$.*

1. $\deg(f + g) \leq \max\{\deg f, \deg g\}$.
2. $\deg(fg) \leq \deg f + \deg g$.
3. *If R has no zero divisors, $\deg(fg) = \deg f + \deg g$.*
4. *If $n = 1$ and the leading coefficient of f or g is not a zero divisor, then $\deg(fg) = \deg f + \deg g$.*

Theorem 3.3 (The division algorithm). *Let R be a ring and $f, g \in R[x]$ nonzero polynomials such that the leading coefficient of g is a unit in R . Then there exist unique polynomials $q, r \in R[x]$ such that*

$$f = qg + r \text{ and } \deg r < \deg g$$

Corollary 3.4 (Remainder Theorem). *Let R be a ring with identity and*

$$f(x) = \sum_{i=0}^n a_i x^i \in R[x].$$

For any $c \in R$ there exists a unique $q(x) \in R[x]$ such that $f(x) = q(x)(x - c) + f(c)$.

Corollary 3.5. *If F is a field, then the polynomial ring $F[x]$ is a Euclidean domain, whence $F[x]$ is a principal ideal domain and a unique factorization domain.*

Definition 3.6. *Let R be a subring of a commutative ring S , $c_1, c_2, \dots, c_n \in S$ and $f = \sum_{i=0}^m a_i x_1^{k_{i1}} \cdots x_n^{k_{in}} \in R[x_1, \dots, x_n]$ a polynomial such that $f(c_1, c_2, \dots, c_n) = 0$. Then $(c_1, c_2, \dots, c_n)$ is said to be a root or zero of f (or a solution of the polynomial equation $f(x_1, \dots, x_n) = 0$).⁴*

Theorem 3.7. *Let R be a commutative ring with identity and $f \in R[x]$. Then $c \in R$ is a root of f if and only if $x - c$ divides f .*

Theorem 3.8. *If D is an integral domain contained in an integral domain E and $f \in D[x]$ has degree n , then f has at most n distinct roots in E .*

Definition 3.9. *Let D be an integral domain and $f \in D[x]$. If $c \in D$ and c is a root of f , then there is a greatest integer m ($0 \leq m \leq \deg f$) such that*

$$f(x) = (x - c)^m g(x)$$

where $g(x) \in R[x]$ and $x - c \nmid g(x)$. The integer m is called the **multiplicity** of the root c of f . If c has multiplicity 1, c is said to be a **simple root**. If c has multiplicity $m > 1$, c is called a multiple root.

Theorem 3.10. *Let D be an integral domain which is a subring of an integral domain E . Let $f \in D[x]$ and $c \in E$.*

1. c is a multiple root of f if and only if $f(c) = 0$ and $f'(c) = 0$.
2. If D is a field and f is relatively prime to f' , then f has no multiple roots in E .
3. If D is a field, f is irreducible in $D[x]$ and E contains a root of f , then f has no multiple roots in E if and only if $f' \neq 0$.

§3.1 Over U.F.D

Definition 3.11. *Let D be a unique factorization domain and*

$$f = \sum_{i=0}^n a_i x^i$$

a nonzero polynomial in $D[x]$. A greatest common divisor of the coefficients $a_0, a_1, \dots, a_n$ is called a **content** of f and is denoted $\text{Cont}(f)$.

If $f \in D[x]$ and $\text{Cont}(f)$ is a unit in D , then f is said to be **primitive**. Clearly for any polynomial $g \in D[x]$, $g = \text{Cont}(g)g_1$ with g_1 primitive.

Theorem 3.12 (Gauss). *If D is a unique factorization domain and $f, g \in D[x]$, then $C(fg) \approx C(f)C(g)$. In particular, the product of primitive polynomials is primitive.*

Proof. $f = \text{Cont}(f)f_1$ and $g = \text{Cont}(g)g_1$ with f_1, g_1 primitive. Consequently, $\text{Cont}(g) = C(\text{Cont}(f)f_1 \text{Cont}(g)g_1) \approx \text{Cont}(f) \text{Cont}(g)C(f_1g_1)$. Hence it suffices to prove that f_1g_1 is primitive (that is, $C(f_1g_1)$ is a unit).

If $f_1 = \sum_{i=0}^n a_i x^i$ and $g_1 = \sum_{j=0}^m b_j x^j$, then $f_1g_1 = \sum_{k=0}^{m+n} c_k x^k$ with $c_k = \sum_{i+j=k} a_i b_j$. If f_1g_1 is not primitive, then there exists an irreducible element p in R such that $p \mid c_k$ for all k . Since $C(f_1)$ is a unit $p \nmid C(f_1)$, whence there is a least integer s such that

$$p \mid a_i \text{ for } i < s \text{ and } p \nmid a_s.$$

Similarly there is a least integer t such that

$$p \mid b_j \text{ for } j < t \text{ and } p \nmid b_t.$$

Since p divides $c_{s+t} = a_0 b_{s+t} + \cdots + a_{s-1} b_{t+1} + a_s b_t + a_{s+1} b_{t-1} + \cdots + a_{s+t} b_0$, p must divide $a_s b_t$. Since every irreducible element in D is prime, $p \mid a_s$ or $p \mid b_t$. This is a contradiction. Therefore f_1g_1 is primitive. $\square$

Corollary 3.13. *Let D be a unique factorization domain with quotient field F*

1. *If f is a primitive polynomial of positive degree in $D[x]$, then f is irreducible in $D[x]$ if and only if f is irreducible in $F[x]$.*
2. *If f and g are primitive polynomials in $D[x]$. Then f and g are associates in $D[x]$ if and only if they are associates in $F[x]$.*

Theorem 3.14. *Let D be a unique factorization domain with quotient field F and let $f = \sum_{i=0}^n a_i x^i \in D[x]$. If $u = c/d \in F$ with c and d relatively prime, and u is a root of f , then c divides a_0 and d divides a_n .*

Theorem 3.15. *If D is a Unique Factorization Domain, then so $D[x]$.*

Proof. We have indicated above that $R[x]$ a Unique Factorization Domain forces R to be a Unique Factorization Domain. Suppose conversely that R is a Unique Factorization Domain, F is its field of fractions and $p(x)$ is a nonzero element of $R[x]$. Let d be

the greatest common divisor of the coefficients of $p(x)$, so that $p(x) = dp'(x)$, where the g.c.d. of the coefficients of $p'(x)$ is 1. Such a factorization of $p(x)$ is unique up to a change in d (so up to a unit in R), and since d can be factored uniquely into irreducibles in R (and these are also irreducibles in the larger ring $R[x]$), it suffices to prove that $p'(x)$ can be factored uniquely into irreducibles in $R[x]$. Thus we may assume that the greatest common divisor of the coefficients of $p(x)$ is 1. We may further assume $p(x)$ is not a unit in $R[x]$, i.e., $\text{degree } p(x) > 0$.

Since $F[x]$ is a Unique Factorization Domain, $p(x)$ can be factored uniquely into irreducibles in $F[x]$. By Gauss' Lemma, such a factorization implies there is a factorization of $p(x)$ in $R[x]$ whose factors are F -multiples of the factors in $F[x]$. Since the greatest common divisor of the

coefficients of $p(x)$ is 1, the g.c.d. of the coefficients in each of these factors in $R[x]$ must be 1. By Corollary 6, each of these factors is an irreducible in $R[x]$. This shows that $p(x)$ can be written as a finite product of irreducibles in $R[x]$.

The uniqueness of the factorization of $p(x)$ follows from the uniqueness in $F[x]$. Suppose

$$p(x) = q_1(x) \cdots q_r(x) = q'_1(x) \cdots q'_s(x)$$

are two factorizations of $p(x)$ into irreducibles in $R[x]$. Since the g.c.d. of the coefficients of $p(x)$ is 1, the same is true for each of the irreducible factors above in particular, each has positive degree. By Corollary 6, each $q_i(x)$ and $q'_j(x)$ is an irreducible in $F[x]$. By unique factorization in $F[x]$, $r = s$ and, possibly after rearrangement, $q_i(x)$ and $q'_i(x)$ are associates in $F[x]$ for all $i \in \{1, \dots, r\}$. It remains to show they are associates in $R[x]$. Since the units of $F[x]$ are precisely the elements of $F^\times$ we need to consider when $q(x) = \frac{a}{b}q'(x)$ for some $q(x), q'(x) \in R[x]$ and nonzero elements a, b of R , where the greatest common divisor of the coefficients of each of $q(x)$ and $q'(x)$ is 1. In this case $bq(x) = aq'(x)$; the g.c.d. of the coefficients on the left hand side is b and on the right hand side is a . Since in a Unique Factorization Domain the g.c.d. of the coefficients of a nonzero polynomial is unique up to units, $a = ub$ for some unit u in R . Thus $q(x) = uq'(x)$ and so $q(x)$ and $q'(x)$ are associates in R as well. This completes the proof. $\square$

Corollary 3.16. *If R is a Unique Factorization Domain, then a polynomial ring in an arbitrary number of variables with coefficients in R is also a Unique Factorization Domain.*

Theorem 3.17 (Eisenstein's Criterion). *Let D be a unique factorization domain with quotient field F . If $f = \sum_{i=0}^n a_i x^i \in D[x]$, $\deg f \geq 1$ and p is an irreducible element of D such that*

$$(i) \quad p \nmid a_n$$

$$(ii) \quad p \mid a_i \text{ for } i = 0, 1, \dots, n-1$$

$$(iii) \quad p^2 \nmid a_0$$

then f is irreducible in $F[x]$. If f is primitive, then f is irreducible in $D[x]$.

Part III

Modules Theory

Chapter VIII

Modules

§1 Basic Definition

Definition 1.1. Let R be a ring. A **left R -module** is an additive abelian group M together with a function $R \times M \rightarrow M$ such that for all $r, s \in R$ and $a, b \in M$:

$$(i) \quad r(a + b) = ra + rb.$$

$$(ii) \quad (r + s)a = ra + sa$$

$$(iii) \quad r(sa) = (rs)a$$

$$(iv) \quad 1_R a = a \text{ for all } a \in M$$

then M is said to be a **R -module**. If R is a division ring, then a R -module is called a (left) **vector space**.

Remark. If R has no identity or (iv) fails, we call M **non-unital module** or **pseudo-module**. In the vast majority of cases, the objects we study are modules.

Definition 1.2. Let R be a ring, M an R -module and N a nonempty subset of M . Then N is a **submodule** of M provided that $N < M$ and $rb \in N$ for all $r \in R, b \in N$. A submodule of a vector space over a division ring is called a **subspace**.

Proposition 1.3. Let M be an R -module.

1. If N is a nonempty subset of M , then N is a submodule of M if and only if for all $x, y \in N$ and $r \in R, x - y \in N$ and $rx \in N$
2. If $\{N_\alpha\}$ is a chain of submodules, then so $\bigcup N_\alpha$.
3. If $\{N_\alpha\}$ is a family of submodules, then so $\bigcap N_\alpha$.

§1.1 Submodule generated by sets

Definition 1.4. Let M be a R -module.

1. If X is a subset of M , then the intersection of all submodules of M containing X is called the **submodule generated by X** (or spanned by X).
2. If X is finite, and X generates the submodule N , then N is said to be **finitely generated** and X spans N . If $X = \{a\}$, then the submodule generated by X is called the **cyclic module generated by a** .
3. if $\{B_i : i \in I\}$ is a family of submodules of M , then the submodule generated by $X = \bigcup_{i \in I} B_i$ is called the **sum of the modules B_i** , denoted by $\sum_{i \in I} B_i$.
4. If I is a left ideal of R and S is a nonempty subset of M . Then

$$IS = \left\{ \sum_{i=1}^n r_i a_i : r_i \in I; a_i \in S; n \in \mathbb{N}^* \right\}$$

is a submodule of M . Similarly if $a \in M$, then $Ia = \{ra \mid r \in I\}$ is a submodule of M .

Theorem 1.5. Let R be a ring, M an R -module, X a subset of M , $\{M_\alpha\}$ a family of submodules of M .

1. The submodule generated by X is

$$RX = \left\{ \sum_{i=1}^s r_i a_i : s \in \mathbb{Z}_{\geq 1}; a_i \in X; r_i \in R \right\}$$

2. The submodule generated by the family $\{M_\alpha\}$ consists of all finite sums that is

$$\sum M_\alpha = \{x_{i_1} + \cdots + x_{i_n} : n \in \mathbb{Z}_{\geq 1}; x_{i_k} \in M_{\alpha_k}\}$$

§1.2 Quotient module and homomorphism

Definition 1.6. Let M and N be modules over a ring R .

1. A function $f : M \rightarrow N$ is an **R -module homomorphism** provided that for all $x, y \in M$ and $r \in R$:

$$f(x + y) = f(x) + f(y) \text{ and } f(rx) = rf(x)$$

If R is a division ring, then an R -module homomorphism is called a **linear transformation**.

2. Four submodules $\text{Ker } f = f^{-1}(0)$, $\text{Im}(f) = f(M)$, $\text{Coker}(f) = N / \text{Im}(f)$, $\text{Coim}(f) = M / \text{Ker}(f)$

3. define $\text{Hom}_R(M, N)$ to be the set of all R -module homomorphisms from M into N , which forms a R -module.

Proposition 1.7. *The following conditions of module-homomorphism $f : M \rightarrow N$ are equivalent*

1. f is injective
2. $\text{Ker}(f) = 0$
3. f is monomorphism ($fg = fh \Rightarrow g = h$)
4. $fg = 0 \Rightarrow g = 0$

Definition 1.8. *Let N be a submodule of a module M over a ring R . Then the quotient group M/N is an R -module with the action of R on M/N given by:*

$$r(x + N) = rx + N \quad \text{for all } r \in R, x \in M$$

*called quotient module. The map $\pi : M \rightarrow M/N$ given by $x \mapsto x + N$ is an R -module epimorphism with kernel N . The map π is called the **canonical epimorphism (or projection)**.*

Theorem 1.9. *Let R be a ring, $f : M \rightarrow N$ an R -module homomorphism and L is a submodule of $\text{Ker } f$, then*

1. *There is a unique R -module homomorphism*

$$\bar{f} : M/L \rightarrow N \quad \text{such that } \bar{f}(x + L) = f(x) \text{ for all } x \in M$$

with $\text{Im } \bar{f} = \text{Im}\{f\}$ and $\text{Ker } \bar{f} = \text{Ker } f/L$.

2. $\bar{f}$ is an R -module isomorphism if and only if f is an R -module epimorphism and $C = \text{Ker } f$. In particular, $M/\text{Ker } f \cong \text{Im}\{f\}$.

Corollary 1.10. *If R is a ring and M' is a submodule of the R -module M and N' a submodule of the R -module N and $f : M \rightarrow N$ is an R -module homomorphism such that $f(A') \subset B'$, then*

- (1) f induces an R -module homomorphism $\bar{f} : A/A' \rightarrow B/B'$ given by $a + A' \mapsto f(a) + B'$. $\bar{f}$ is an R -module isomorphism if and only if $\text{Im}\{f\} + B' = B$ and $f^{-1}(B') \subset A'$.

In particular if f is an epimorphism such that $f(A') = B'$ and $\text{Ker } f \subset A'$, then $\bar{f}$ is an R -module isomorphism.

Theorem 1.11. *Let B and C be submodules of a module A over a ring R .*

- (1) *There is an R -module isomorphism $B/(B \cap C) \cong (B + C)/C$;*
- (2) *if $C \subset B$, then B/C is a submodule of A/C , and there is an R -module isomorphism $(A/C)/(B/C) \cong A/B$.*

Theorem 1.12. *Let R be a ring and N is a submodule of an R -module M , then*

1. There is a one-to-one correspondence between the set of all submodules of M containing N and the set of all submodules of M/N , given by $L \mapsto L/N$.
2. Hence every submodule of M/N is of the form L/N , where L is a submodule of M which contains N .

§1.3 Annihilator

Definition 1.13. Let R be a ring and M an left R -module. If X be a subset of M

1. The left ideal

$$\text{Ann}(X) = \{r \in R \mid rX = 0\}$$

is called the **annihilator** of X in R .

2. If $\text{Ann}(u) \neq 0$, u is called to be **torsion element**.

If $\text{Ann}(u) = 0$, u is called to be **free-torsion element**.

3. If $\text{Ann}(u) = 0$ for all $u \in M$, then M is called to be a **torsion-free** module.

If $\text{Ann}(u) \neq 0$ for all $u \in M$, then M is called to be a **torsion module**.

4. If R is a integral domain, the set $T(M)$ consists of all torsion element is called the **torsion submodule** of M

Proposition 1.14. Let N, N_1, N_2 be submodule of M .

1. $\text{Ann}(N_1 + N_2) = \text{Ann}(N_1) + \text{Ann}(N_2)$
2. $\text{Ann}(M/N) = (N : M)$
3. $(N_1 : N_2) = (N_1 : (N_1 + N_2)) = \text{Ann}((N_1 + N_2)/N_1)$

Definition 1.15. The module M is said to be a **faithful** R -module if $\text{Ann}(M) = 0$.

Remark. It's obvious that M is a faithful $R/\text{Ann}(M)$ -module

§2 Modules Category

We only consider left module in this section.

§2.1 Direct Products and Direct Sums

Definition 2.1. Let R be a ring and $\{M_i : i \in I\}$ a nonempty family of left R -modules, $\prod_{i \in I} M_i$ the direct product of the abelian groups M_i , and $\bigoplus_{i \in I} M_i$ the direct sum of the abelian groups M_i .

1. $\prod_{i \in I} M_i$ is an left R -module with the action of R given by $r \{a_i\} = \{ra_i\}$.

2. $\bigoplus_{i \in I} M_i$ is a submodule of $\prod_{i \in I} M_i$.
3. For each $k \in I$, the canonical projection $\pi_k : \prod M_i \rightarrow M_k$ is an left R -module epimorphism.
4. For each $k \in I$, the canonical injection $\iota_k : M_k \rightarrow \bigoplus M_i$ is an left R -module monomorphism.

The ring $\prod_{i \in I} M_i$ is called the **(external) direct product** of the family of R -modules $\{M_i \mid i \in I\}$ and $\bigoplus_{i \in I} M_i$ is its **(external) direct sum**. The maps π_k are called the **canonical projections** and ι_k are called **canonical injections**.

Theorem 2.2. Let R be a ring.

1. If $\{M_i \mid i \in I\}$ a family of R -modules, M an R -module, and $\{f_i : M \rightarrow M_i \mid i \in I\}$ a family of R -module homomorphisms, then there is a unique R -module homomorphism $f : M \rightarrow \prod_{i \in I} M_i$ such that the diagram

$$\begin{array}{ccc}
 M & \xrightarrow{\exists! f} & \prod_{i \in I} M_i \\
 \downarrow f_i & \nearrow \pi_i & \\
 M_i & &
 \end{array}$$

is commutative for all $i \in I$. $\prod_{i \in I} M_i$ is uniquely determined up to isomorphism by this property. In other words, $\prod_{i \in I} M_i$ is a product in the category of left R -modules.

2. If $\{N_i\}_{i \in I}$ is a family of R -modules, N an R -module, and $g_i \in \text{Hom}_R(N_i, N) \mid i \in I$, then there is a unique R -module homomorphism $g : \bigoplus_{i \in I} N_i \rightarrow N$ such that the diagram

$$\begin{array}{ccc}
 \bigoplus_{i \in I} N_i & \xrightarrow{\exists! g} & N \\
 \downarrow \iota_i & \nearrow g_i & \\
 N_i & &
 \end{array}$$

is commutative for all $i \in I$. $\bigoplus_{i \in I} N_i$ is uniquely determined up to isomorphism by this property. In other words, $\bigoplus_{i \in I} N_i$ is a coproduct in the category of R -modules.

Remark. It follows that

$$\prod \text{Hom}_R(A, A_i) \cong \text{Hom}_R\left(A, \prod A_i\right)$$

$$\prod \text{Hom}_R(A_i, A) \cong \text{Hom}_R\left(\bigoplus A_i, A\right)$$

Theorem 2.3. *Let R be a ring and $M, M_1, M_2, \dots, M_n, R$ -modules. Then $M \cong M_1 \oplus M_2 \oplus \dots \oplus M_n$ if and only if for each $i = 1, 2, \dots, n$ there are R -module homomorphisms $\pi_i : M \rightarrow M_i$ and $\iota_i : M_i \rightarrow M$ such that*

- (i) $\pi_i \iota_i = 1_{M_i}$ for $i = 1, 2, \dots, n$
- (ii) $\pi_j \iota_i = 0$ for $i \neq j$
- (iii) $\iota_1 \pi_1 + \iota_2 \pi_2 + \dots + \iota_n \pi_n = 1_M$.

Theorem 2.4. *Let R be a ring and $\{M_i : i \in I\}$ a family of submodules of an R -module M such that*

- (i) $M = \sum_i M_i$
- (ii) for each $k \in I$, $M_k \cap \sum_{i \neq k} M_i = 0$

*Then there is an isomorphism $M \cong \bigoplus_{i \in I} M_i$. The module M is said to be the **internal direct sum** of $\{M_i : i \in I\}$.*

Corollary 2.5. *Let R be a ring and $\{R_i : i \in I\}$ a family of subrings of an R such that*

- (i) $R = \sum_i R_i$
- (ii) for each $k \in I$, $R_k \cap \sum_{i \neq k} R_i = 0$

Then there is an isomorphism $M \cong \bigoplus_{i \in I} R_i$.

§2.2 Free Modules

Definition 2.6. *Let R be a ring and M an left R -module*

1. *A subset X of M is said to be **linearly independent** provided that for some finite distinct $x_1, \dots, x_n \in X$ and $r_i \in R$.*

$$r_1 x_1 + r_2 x_2 + \dots + r_n x_n = 0 \Rightarrow r_i = 0 \text{ for every } i$$

*Conversely, a set that is not linearly independent is said to be **R-linearly dependent**.*

2. *The $\{u_i\}_{i \in I}$ is called to be a **maximal linearly independent subset** of M provided that it is not contained in any larger linearly independent subset of M .*
3. *A linearly independent subset that spans M is called a **(Hamel) basis** of M .*

Remark. *A basis must be a maximal linearly independent subset.*

Definition 2.7. Let R be a ring and X a nonempty set. An left R -module F is called a **free module** on X if F is a free object on X in the category of all left R -modules, i.e. there is a function $\iota : X \rightarrow F$ such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{\iota} & F(X) \\ & \searrow f & \downarrow \exists! \tilde{f} \\ & & M \end{array}$$

is commutative for any left R -module M and function $f : X \rightarrow M$ there is a unique R -module homomorphism $\tilde{f} : F(X) \rightarrow M$ with $\tilde{f}\iota = f$.

Theorem 2.8. Let R be a ring. The following conditions on a R -module F are equivalent:

1. F has a nonempty basis;
2. F is the internal direct sum of a family of cyclic R -modules, each of which is isomorphic as a left R -module to R ,
3. F is R -module isomorphic to a direct sum of copies of the left R -module R

Corollary 2.9. .

1. Every module M over a ring R is the homomorphic image of a free R -module F .
2. M is a finitely generated $\Leftrightarrow M$ is isomorphic to a quotient of free module R^n for some integer $n > 0$.

Dimension and invariant dimension property

Definition 2.10. Let R be a ring. If any two bases of free R -module F have the same cardinality, then R is said to have the **invariant dimension property** and the cardinal number of any basis of F is called the **dimension** or **rank** of F .

Proof of invariant dimension property

Theorem 2.11. Let R be a ring and F a free R -module with an infinite basis X . Then every basis of F has the same cardinality as X .

Proof. Step 1. If Y is another basis of F , then we claim that Y is infinite. Suppose on the contrary that Y were finite. Since Y generates F and every element of Y is a linear combination of a finite number of elements of X , it follows that there is a finite subset $\{x_1, \dots, x_m\}$ of X , which generates Y , thus generates F . Since X is infinite, there exists

$$x \in X - \{x_1, \dots, x_m\}$$

Then for some $r_i \in R$, $x = r_1x_1 + \dots + r_mx_m$, which contradicts the linear independence of X . Therefore, Y is infinite.

Step 2. Let $K(Y)$ be the set of all finite subsets of Y . Define a map

$$f : X \rightarrow K(Y)$$

by

$$x \mapsto \{y_1, \dots, y_n\}$$

where $x = r_1y_1 + \dots + r_ny_n$ and $r_i \neq 0$ for all i . Since Y is a basis, the y_i are uniquely determined and f is a well-defined function.

Step 3. If $\text{Im } f$ were finite, then

$$\bigcup_{S \in \text{Im } f} S$$

would be a finite subset of Y that would generate X and hence F . This leads to a contradiction of the linear independence of Y as in the preceding paragraph. Hence $\text{Im } f$ is infinite.

Step 4. Next we show that $f^{-1}(T)$ is a finite subset of X for every $T \in \text{Im } f \subset K(Y)$. If $x \in f^{-1}(T)$, then x is contained in the submodule $\langle T \rangle$ of F generated by T ; that is,

$$f^{-1}(T) \subset \langle T \rangle$$

Since T is finite and each $y \in T$ is a linear combination of a finite number of elements of X , there is a finite subset S of X such that $\langle T \rangle \subset \langle S \rangle$. Thus $x \in f^{-1}(T)$ implies $x \in \langle S \rangle$ and x is a linear combination of elements of S . Since $x \in X$ and $S \subset X$, this contradicts the linear independence of X unless $x \in S$. Therefore, $f^{-1}(T) \subset S$, whence $f^{-1}(T)$ is finite.

Step 5. Verify that the sets $f^{-1}(T)$ form a partition of X ,

$$X = \bigsqcup_{T \in \text{Im } f} f^{-1}(T)$$

For each $T \in \text{Im } f$, order the elements of $f^{-1}(T)$, say $x_1, \dots, x_n$, and define an injective map

$$g_T : f^{-1}(T) \rightarrow \text{Im } f \times N \quad \text{by } x_k \mapsto (T, k).$$

It follows that the map $X \rightarrow \text{Im } f \times N$ defined by $x \mapsto g_T(x)$, where $x \in f^{-1}(T)$, is a well-defined injective function, whence $|X| \leq |\text{Im } f \times N|$. Therefore

$$|X| \leq |\text{Im } f \times N| = |\text{Im } f| \leq |K(Y)| = |Y|$$

since $|\text{Im } f|$ is infinite.

Step 6. Interchanging X and Y in the preceding argument shows that $|Y| \leq |X|$. Therefore $|Y| = |X|$ by the Schroeder-Bernstein Theorem. $\square$

Lemma 2.12. *Let R be a ring, $I (\neq R)$ an ideal of R , F a free R -module with basis X . Then F/IF is a free R/I -module with basis $\pi(X)$ and $|\pi(X)| = |X|$ where $\pi : F \rightarrow F/IF$ is the canonical epimorphism.*

Proof. If $u + IF \in F/IF$, then $u = \sum_{j=1}^n r_j x_j$ with $r_j \in R, x_j \in X$ since $u \in F$ and X is a basis of F . Consequently,

$$u + IF = \sum_j (r_j x_j + IF) = \sum_j (r_j + I) (x_j + IF) = \sum_j (r_j + I) \pi(x_j)$$

whence $\pi(X)$ generates F/IF as an R/I -module.

On the other hand, if $\sum_{k=1}^m (r_k + I) \pi(x_k) = 0$ with $r_k \in R$ and $x_1, \dots, x_m$ distinct elements of X , then

$$0 = \sum_k (r_k + I) \pi(x_k) = \sum_k (r_k + I) (x_k + IF) = \sum_k r_k x_k + IF$$

whence $\sum_k r_k x_k \in IF$. Thus $\sum_k r_k x_k = \sum_j s_j u_j$ with $s_j \in I, u_j \in F$. Since each u_j is a linear combination of elements of X and I is an ideal, $\sum_j s_j u_j$ is a linear combination of elements of X with coefficients in I . Consequently,

$$\sum_{k=1}^m r_k x_k = \sum_j s_j u_j = \sum_{t=1}^d c_t y_t$$

with $c_t \in I, y_t \in X$. The linear independence of X implies that $r_k \in I$ for every k . Hence $r_k + I = 0$ in R/I for every k and $\pi(X)$ is linearly independent over R/I . Thus F/IF is a free R/I -module with basis $\pi(X)$.

Finally if $x, x' \in X$ and $\pi(x) = \pi(x')$ in F/IF , then

$$0 = (1_R + I) \pi(x) - (1_R + I) \pi(x') = (1_R + I) (x + IF) - (1_R + I) (x' + IF)$$

If $x \neq x'$, the preceding argument implies that $1_R + I = 0$ in R/I , which contradicts the fact that $I \neq R$. Therefore, $x = x'$ and the map $\pi : X \rightarrow \pi(X)$ is a bijection, whence $|X| = |\pi(X)|$. $\square$

Theorem 2.13. .

1. Division ring has the invariant dimension property.
2. Let $f : R \rightarrow S$ be a nonzero epimorphism of rings. If S has the invariant dimension property, then so does R .
3. If R is a ring that has a homomorphic image which is a division ring, then R has the invariant dimension property.
4. Every commutative ring has the invariant dimension property.

Proof. (2) Let $I = \text{Ker } f$; then $S \cong R/I$. Let X and Y be two bases of the free unitary R -module F and $\pi : F \rightarrow F/IF$ the canonical epimorphism. By Lemma F/IF is a free R/I -module (and hence a free unitary S -module) with bases $\pi(X)$ and $\pi(Y)$ such that $|X| = |\pi(X)|, |Y| = |\pi(Y)|$.

Since S has the invariant dimension property, $|\pi(X)| = |\pi(Y)|$. Therefore, $|X| = |Y|$ and R has the invariant dimension property.

(3) It is obvious from (1) and (2)

(4) Consider the maximal ideal $\mathfrak{m}$ in R and canonical projective

$$R \rightarrow R/\mathfrak{m}$$

□

Theorem 2.14. *Let R be a ring and M a module over R . Let I be a non-empty set, and let $\{x_i\}_{i \in I}$ be a basis of M . Let N be an R -module, and let $\{y_i\}_{i \in I}$ be a family of elements of N . Then there exists a unique homomorphism $f : M \rightarrow N$ such that $f(x_i) = y_i$ for all i .*

§2.3 Vector Space

Theorem 2.15. *Let D be a division ring.*

1. *A maximal linearly independent subset X of a vector space V over a division ring D is a basis of V .*
2. *Basis Extension Theorem. Let V be a vector space, T spans V and S be a subset of T which is linearly independent. Then there exists a basis B of V such that $S \subset B \subset T$.*
3. *Every vector space V over a division ring D has a basis and is therefore a free D -module.*
4. *Let V be a vector space over D . Then two bases of V over D have the same cardinality.*

Definition 2.16. *If V is a finitely generated D -module the cardinality of any basis is called the **dimension of V** and is denoted by $\dim_D V$, or just $\dim V$, and V is said to be **finite dimensional** over D . If V is not finitely generated, V is said to be **infinite dimensional** (written $\dim V = \infty$).*

Theorem 2.17. *Let W be a subspace of a vector space V over a division ring D .*

1. $\dim_D W \leq \dim_D V$
2. *if $\dim_D W = \dim_D V < \infty$, then $W = V$*
3. $\dim_D V = \dim_D W + \dim_D(V/W)$.
4. *If V_1 and V_2 are finite dimensional subspaces of a vector space over D , then*

$$\dim_D V_1 + \dim_D V_2 = \dim_D(V_1 \cap V_2) + \dim_D(V_1 + V_2)$$

Corollary 2.18. *If $f : V \rightarrow V'$ is a linear transformation of vector spaces over a division ring D , then there exists a basis X of V such that $X \cap \text{Ker } f$ is a basis of $\text{Ker } f$ and $X \cap f^{-1}(\text{Im } f)$ is a basis of $\text{Im } f$. In particular,*

$$\dim_D V = \dim_D(\text{Ker } f) + \dim_D(\text{Im } f)$$

Theorem 2.19. *Let R, S, T be division rings such that $R \subset S \subset T$. Then*

$$\dim_R T = (\dim_S T) (\dim_R S)$$

Furthermore, $\dim_R T$ is finite if and only if $\dim_S T$ and $\dim_R S$ are finite.

§2.4 Pullbacks and Pushout

Definition 2.20. *Let R be a ring and in $R\text{-Mod}$.*

1. *Given a diagram*

$$A_1 \xrightarrow{f_1} B \xleftarrow{f_2} A_2$$

*The **pullback** of f_1 and f_2 is the submodule*

$$P = \{(a_1, a_2) \in A_1 \oplus A_2 \mid f_1(a_1) = f_2(a_2)\}$$

of the direct product $A_1 \oplus A_2$ together with the canonical projections $\pi_1 : P \rightarrow A_1$ and $\pi_2 : P \rightarrow A_2$. That is,

$$\begin{array}{ccccc} X & & & & \\ & \searrow \exists! u & & \searrow q_2 & \\ & & P & \xrightarrow{\pi_2} & A_2 \\ & \searrow q_1 & \downarrow \pi_1 & & \downarrow f_2 \\ & & A_1 & \xrightarrow{f_1} & B \end{array}$$

2. *Given a diagram*

$$B_1 \xleftarrow{f_1} A \xrightarrow{f_2} B_2$$

*The **pushout** of f_1 and f_2 is the quotient module*

$$P = (B_1 \oplus B_2)/K$$

where K is the submodule of $B_1 \oplus B_2$ generated by all elements of the form $(f_1(a), -f_2(a))$ with $a \in A$, together with the canonical injections $\iota_1 : B_1 \rightarrow P$ and $\iota_2 : B_2 \rightarrow P$. That is,

$$\begin{array}{ccccc} A & \xrightarrow{f_2} & B_2 & & \\ f_1 \downarrow & & \downarrow \iota_2 & \searrow & \\ B_1 & \xrightarrow{\iota_1} & P & \xrightarrow{\quad} & Y \\ & \searrow & \downarrow & \nearrow & \\ & & & & \end{array}$$

§3 Tensor Products

§3.1 Basic definition

Definition 3.1. Let R be a ring, A_R be a right module and ${}_R B$ a left module. Let F be the free abelian group on the set $A \times B$. Let K be the subgroup of F generated by all elements of the following forms :

$$(a + a', b) - (a, b) - (a', b), (a, b + b') - (a, b) - (a, b'), (ar, b) - (a, rb)$$

The quotient group F/K is called the **tensor product** of A_R and ${}_R B$; it is denoted $A \otimes_R B$. The coset $(a, b) + K$ of the element (a, b) in F is denoted $a \otimes b$; the coset of $(0, 0)$ is denoted 0 .

Theorem 3.2. Let R and S be rings and ${}_S A_R, {}_R B, C_R, {}_R D_S$ bimodules as indicated.

1. $A \otimes_R B$ is a left S -module such that

$$s(a \otimes b) := sa \otimes b$$

for all $s \in S, a \in A, b \in B$.

2. $C \otimes_R D$ is a right S -module such that $(c \otimes d)s = c \otimes ds$ for all $c \in C, d \in D, s \in S$.

Remark. An important special case occurs when R is a commutative ring and hence every R -module A is an R - R bimodule. In this case $A \otimes_R B$ is also an R - R bimodule with

$$r(a \otimes b) = ra \otimes b = ar \otimes b = a \otimes rb = a \otimes br = (a \otimes b)r$$

for all $r \in R, a \in A, b \in B$.

Definition 3.3. If A_R and ${}_R B$ are modules over a ring R , and C is an abelian group, then a **middle linear (or balanced) map** from $A \times B$ to C is a function $f : A \times B \rightarrow C$ such that for all $a, a_i \in A, b, b_i \in B$, and $r \in R$:

$$f(a_1 + a_2, b) = f(a_1, b) + f(a_2, b)$$

$$f(a, b_1 + b_2) = f(a, b_1) + f(a, b_2)$$

$$f(ar, b) = f(a, rb)$$

Remark. The map $i : A \times B \rightarrow A \otimes_R B$ given by $(a, b) \mapsto a \otimes b$ is a middle linear map, that is, for all $a, a_i \in A, b, b_i \in B$, and $r \in R$

$$(a_1 + a_2) \otimes b = a_1 \otimes b + a_2 \otimes b$$

$$a \otimes (b_1 + b_2) = a \otimes b_1 + a \otimes b_2$$

$$ar \otimes b = a \otimes rb$$

The map i is called the **canonical middle linear map**.

Theorem 3.4 (Universal property of canonical middle linear map). *Let A_R and ${}_R B$ be modules over a ring R , and let C be an abelian group. If $g : A \times B \rightarrow C$ is a middle linear map, then there exists a unique group homomorphism $\bar{g} : A \otimes_R B \rightarrow C$ such that*

$$\begin{array}{ccc} A \times B & \xrightarrow{g} & C \\ \downarrow i & \nearrow \exists! \bar{g} & \\ A \otimes_R B & & \end{array}$$

is commutative. This proves that $i : A \times B \rightarrow A \otimes_R B$ is a universal object in the category of all middle linear maps on $A \times B$, whence $A \otimes_R B$ is uniquely determined up to isomorphism.

Definition 3.5. *Let A, B, C be modules over a commutative ring R . A **bilinear map** from $A \times B$ to C is a function $f : A \times B \rightarrow C$ such that for all $a_i \in A, b_i \in B$, and $r \in R$:*

$$\begin{aligned} f(a_1 + a_2, b) &= f(a_1, b) + f(a_2, b) \\ f(a, b_1 + b_2) &= f(a, b_1) + f(a, b_2) \\ f(ra, b) &= rf(a, b) = f(a, rb) \end{aligned}$$

*If A and B are modules over a commutative ring R , then so is $A \otimes_R B$ and the canonical middle linear map $i : A \times B \rightarrow A \otimes_R B$ is easily seen to be bilinear. In this context i is called the **canonical bilinear map**.*

Theorem 3.6 (Universal property of canonical bilinear map). *If A, B, C are modules over a commutative ring R and $g : A \times B \rightarrow C$ is a bilinear map, then there is a unique R -module homomorphism $\bar{g} : A \otimes_R B \rightarrow C$ such that*

$$\begin{array}{ccc} A \times B & \xrightarrow{g} & C \\ \downarrow i & \nearrow \exists! \bar{g} & \\ A \otimes B & & \end{array}$$

where $i : A \times B \rightarrow A \otimes_R B$ is the canonical bilinear map. The module $A \otimes_R B$ is uniquely determined up to isomorphism by this property.

Proposition 3.7. *If $A_R, A'_R, {}_R B$ and ${}_R B'$ are modules over a [resp. commutative] ring R and $f : A \rightarrow A', g : B \rightarrow B'$ are R -module homomorphisms, then there is a unique group [resp. R -module] homomorphism*

$$f \otimes g : A \otimes_R B \rightarrow A' \otimes_R B' \quad \text{such that } a \otimes b \mapsto f(a) \otimes g(b)$$

for all $a \in A, b \in B$.

§3.2 Operation of tensor products

Theorem 3.8. *Let R and S be rings and ${}_S A_R, {}_R B, C_R, {}_R D_S$ (bi)modules as indicated.*

1. *If $f : A \rightarrow A'$ is a homomorphism of S - R bimodules and $g : B \rightarrow B'$ is an R -module homomorphism, then the induced map $f \otimes g : A \otimes_R B \rightarrow A' \otimes_R B'$ is a homomorphism of left S -modules.*
2. *If $h : C \rightarrow C'$ is an R -module homomorphism and $k : D \rightarrow D'$ a homomorphism of R - S bimodules, then the induced map $h \otimes k : C \otimes_R D \rightarrow C' \otimes_R D'$ is a homomorphism of right S -modules.*

Theorem 3.9. *Let R, S be ring. Then*

1. *If $A_R, {}_R B_S, {}_S C$ are (bi)modules, then there is an abelian group isomorphism.*

$$(A \otimes_R B) \otimes_S C \cong A \otimes_R (B \otimes_S C)$$

If A is an S - R module, there is a left S -module isomorphism $(A \otimes_R B) \otimes_R C \cong A \otimes_R (B \otimes_R C)$.

2. *Let A and $\{A_i \mid i \in I\}$ right R -modules, B and $\{B_j \mid j \in J\}$ left R -modules. Then there are abelian group isomorphisms:*

$$\left(\bigoplus_{i \in I} A_i \right) \otimes_R B \cong \bigoplus_{i \in I} (A_i \otimes_R B)$$

and

$$A \otimes_R \left(\bigoplus_{j \in J} B_j \right) \cong \bigoplus_{j \in J} (A \otimes_R B_j)$$

Proof. 1. By definition, we have

$$v = \sum_i u_i \otimes c_i = \sum_i \left(\sum_j a_{ij} \otimes b_{ij} \right) \otimes c_i = \sum_i \sum_j [(a_{ij} \otimes b_{ij}) \otimes c_i].$$

Therefore, $(A \otimes_R B) \otimes_S C$ is generated by all elements of the form $(a \otimes b) \otimes c$ ($a \in A, b \in B, c \in C$). Similarly, $A \otimes_R (B \otimes_S C)$ is generated by all $a \otimes (b \otimes c)$ with $a \in A, b \in B, c \in C$.

Verify that the assignment $(\sum_{i=1}^n a_i \otimes b_i, c) \mapsto \sum_{i=1}^n [a_i \otimes (b_i \otimes c)]$ defines an S -middle linear map $(A \otimes_R B) \times C \rightarrow A \otimes_R (B \otimes_S C)$. Therefore, by 3.4 there is a homomorphism

$$\alpha : (A \otimes_R B) \otimes_S C \rightarrow A \otimes_R (B \otimes_S C)$$

with $\alpha[(a \otimes b) \otimes c] = a \otimes (b \otimes c)$ for all $a \in A, b \in B, c \in C$. Similarly there is an homomorphism

$$\beta : A \otimes_R (B \otimes_S C) \rightarrow (A \otimes_R B) \otimes_S C$$

such that $\beta[a \otimes (b \otimes c)] = (a \otimes b) \otimes c$ for all $a \in A, b \in B, c \in C$. Therefore, α and β are isomorphisms.

2. Let ι_k, π_k be the canonical injections and projections of $\bigoplus_{i \in I} A_i$. The family of homomorphisms

$$\iota_k \otimes 1_B : A_k \otimes_R B \rightarrow \left(\bigoplus_{i \in I} A_i \right) \otimes_R B$$

induce a homomorphism

$$\alpha : \bigoplus_{i \in I} (A_i \otimes_R B) \rightarrow \left(\bigoplus_{i \in I} A_i \right) \otimes_R B$$

such that $\alpha(\{a_i \otimes b\}) = \bigoplus (\iota_i(a_i) \otimes b) = (\bigoplus \iota_i(a_i)) \otimes b$ (Note that the summation here is finite.) The assignment $(u, b) \mapsto \{\pi_i(u) \otimes b\}$ defines a middle linear map $(\bigoplus_{i \in I} A_i) \times B \rightarrow \bigoplus_{i \in I} (A_i \otimes_R B)$ and thus induces a homomorphism $\beta : (\bigoplus A_i) \otimes_R B \rightarrow \bigoplus (A_i \otimes_R B)$ such that $\beta(u \otimes b) = \{\pi_i(u) \otimes b\}$. We can show that $\alpha\beta$ and $\beta\alpha$ are the respective identity maps, whence α is an isomorphism. $\square$

Theorem 3.10 (Adjoint Associativity). *Let R and S be rings and $A_R, {}_R B_S, C_S$ modules. Then there is an isomorphism of abelian groups*

$$\alpha : \text{Hom}_S(A \otimes_R B, C) \cong \text{Hom}_R(A, \text{Hom}_S(B, C)),$$

defined for each S -module homomorphism $f : A \otimes_R B \rightarrow C$ by

$$f \mapsto f(- \otimes -)$$

where $f(- \otimes -) : A \rightarrow \text{Hom}_S(B, C)$ is the map defined by $a \mapsto f(a \otimes -)$

§3.3

Theorem 3.11. *If R is a ring and $A_R, {}_R B$ are R -modules, then there are right R -module isomorphisms*

$$A \otimes_R R \cong A$$

and left R -module isomorphisms

$$R \otimes_R B \cong B$$

Theorem 3.12. *Let R be a ring. If A is a right R -module and F is a free left R -module with basis Y . Then every element u of $A \otimes_R F$ may be written uniquely in the form $u = \sum_{i=1}^n a_i \otimes y_i$, where $a_i \in A$ and the y_i are distinct elements of Y .*

Proof. For each $y \in Y$, let A_y be a copy of A and consider the direct sum $\bigoplus_{y \in Y} A_y$. We first

construct an isomorphism

$$\theta : A \otimes_R F \cong \bigoplus_{y \in Y} A_y$$

as follows. Since Y is a basis, $\{y\}$ is a linearly independent set for each $y \in Y$. Consequently, the R -module epimorphism

$$\varphi : R \rightarrow Ry \quad \text{given by } r \mapsto ry$$

is actually an isomorphism. Therefore there is for each $y \in Y$ an isomorphism

$$A \otimes_R Ry \xrightarrow{1_A \otimes \varphi^{-1}} A \otimes_R R \cong A = A_y.$$

Thus by 3.9 and I.8.10 there is an isomorphism $\theta :$

$$A \otimes_R F = A \otimes_R \left(\bigoplus_{y \in Y} Ry \right) \cong \bigoplus_{y \in Y} A \otimes_R Ry \cong \bigoplus_{y \in Y} A_y$$

Verify that for every $a \in A, z \in Y$,

$$\theta(a \otimes z) = \{u_y\} \in \bigoplus A_y$$

where $u_z = a$ and $u_y = 0$ for $y \neq z$; in other words, $\theta(a \otimes z) = \iota_z(a)$, with $\iota_z : A_z \rightarrow \bigoplus A_y$ the canonical injection. Now every nonzero $v \in \bigoplus A_y$ is a finite sum $v = \iota_{y_1}(a_1) + \cdots + \iota_{y_n}(a_n) = \theta(a_1 \otimes y_1) + \cdots + \theta(a_n \otimes y_n)$ with $y_1, \dots, y_n$ distinct elements of Y and a_i uniquely determined nonzero elements of A . It follows that every element of $A \otimes_R F$ (which is necessarily $\theta^{-1}(v)$ for some v) may be written uniquely as $\bigoplus_{i=1}^n a_i \otimes y_i$. $\square$

Corollary 3.13. *If R is a ring with identity and A_R and ${}_R B$ are free R -modules with bases X and Y respectively, then $A \otimes_R B$ is a free (right) R -module with basis $W = \{x \otimes y \mid x \in X, y \in Y\}$ of cardinality $|X||Y|$.*

REMARKS. Since R is an $R - R$ bimodule, so is every direct sum of copies of R . In particular, every free left R -module is also a free right R -module and vice versa. However, it is not true in general that a free (left) R -module is a free object in the category of $R - R$ bimodules (Exercise 12).

Proof. By the proof of Theorem 5.11 and by Theorem 2.1 (for right R -modules) there is a group isomorphism

$$\theta : A \otimes_R B \cong \bigoplus_{y \in Y} A_y = \sum_{y \in Y} A = \sum_{y \in Y} \left(\sum_{x \in X} xR \right).$$

Since B is an $R - R$ bimodule by the remark preceding the proof, $A \otimes_R B$ is a right R -module by Theorem 5.5. Verify that θ is an isomorphism of right R -modules such that $\theta(W)$ is a basis of the free right R -module $\sum_Y (\sum_X xR)$. Therefore, $A \otimes_R B$ is a free right R -module with basis W . Since the elements of W are all distinct by Theorem 5.11, $|W| = |X||Y|$. $\square$

§4 Algebra

Definition 4.1. Let K be a commutative ring. The abelian group $(A, +)$ is a **R -algebra** if

- (i) $(A, +, \cdot)$ is a R -module
- (ii) $(A, +, \times)$ is a ring
- (iii) $r \cdot (a \times b) = (ra)b = a(rb)$ for all $r \in R$ and $a, b \in A$.

If A which, as a ring, is a division ring, is called a **division algebra**.

Remark. Condition (ii) may fail to hold if A does not have an identity or noassociativity. In this case, we say that A is a **nonunital R -algebra** or **noassociativity R -algebra** respectively.

An algebra over a field F that is finite dimensional as a vector space over K is called a **finite dimensional algebra** over F .

Definition 4.2. Let K be a commutative ring with identity 1 and A, B K -algebras.

1. A **subalgebra** of A is a subring of A that is also a K -submodule of A .
2. A (left, right, two-sided) **algebra ideal** of A is a (left, right, two-sided) ideal of the ring A (in this case it is also a K -submodule of A).
3. A homomorphism [resp. isomorphism] of K -algebras $f : A \rightarrow B$ is a ring homomorphism [isomorphism] that is also a K -module homomorphism [isomorphism].

Theorem 4.3 (Tensor product of algebras). Let A and B be algebras over a commutative ring K . Let π be the composition

$$(A \otimes_K B) \otimes_K (A \otimes_K B) \xrightarrow{1_A \otimes \alpha \otimes 1_B} (A \otimes_K A) \otimes_K (B \otimes_K B) \xrightarrow{\pi_A \otimes \pi_B} A \otimes_K B$$

where π_A, π_B are the product maps of A and B respectively. Then $A \otimes_K B$ is a K -algebra with product map $\pi = \pi_A \otimes \pi_B$. The K -algebra $A \otimes_K B$ is called the **tensor product of the K -algebras A and B** .

Proof. note that for generators $a \otimes b$ and $a_1 \otimes b_1$ of $A \otimes_K B$ the product is defined to be

$$(a \otimes b)(a_1 \otimes b_1) = \pi(a \otimes b \otimes a_1 \otimes b_1) = aa_1 \otimes bb_1$$

Thus if A and B have identities $1_A, 1_B$ respectively, then $1_A \otimes 1_B$ is the identity in $A \otimes_K B$. $\square$

§5 Modules over Principal Ideal Domains

Rings are P.I.D and R -module M .

§5.1 Preparatory Lemmas

Theorem 5.1. *Let F be a free module over R and N a submodule of F . Then N is a free R -module and $\text{rank } N \leq \text{rank } F$.*

Proof. Let $\{x_1, x_2, \dots, x_n\}$ be a basis of F . We proceed by induction on n .

If $n = 1$, then $F \cong R$ and $N \cong I$ for some ideal I of R . Since R is a principal ideal domain, $I = (a)$ for some $a \in R$. Therefore, $N \cong R$ or $N = 0$ is free and $\text{rank } N \leq \text{rank } F$.

Suppose the theorem is true for $n - 1$ and let F be generated by n elements. Let $\pi : F \rightarrow R$ be the projection defined by

$$\pi(r_1x_1 + r_2x_2 + \dots + r_nx_n) = r_n$$

for all $r_i \in R$. Then $\pi(N)$ is an ideal of R , whence $\pi(N) = (a)$ for some $a \in R$ since R is a principal ideal domain.

If $\pi(N) = 0$, then $N \subseteq \text{Ker } \pi = \langle x_1, x_2, \dots, x_{n-1} \rangle$. By the induction hypothesis, N is free and $\text{rank } N \leq n - 1 < \text{rank } F$.

If $\pi(N) \neq 0$, let $z \in N$ such that $\pi(z) = a$. For each $y \in N$, there exists $r \in R$ such that $\pi(y) = ra = \pi(rz)$. Thus $y - rz \in \text{Ker } \pi \cap N$. It follows that

$$N = Rz \oplus (\text{Ker } \pi \cap N)$$

Since $\text{Ker } \pi \cap N$ is a submodule of the free module $\text{Ker } \pi = \langle x_1, x_2, \dots, x_{n-1} \rangle$, $\text{Ker } \pi \cap N$ is free and $\text{rank}(\text{Ker } \pi \cap N) \leq n - 1$ by the induction hypothesis. Therefore, N is free and

$$\text{rank } N = 1 + \text{rank}(\text{Ker } \pi \cap N) \leq 1 + (n - 1) = n = \text{rank } F$$

□

Corollary 5.2. *If M is a finitely generated by n elements, then every submodule of M may be generated by m elements with $m \leq n$.*

Corollary 5.3. *A module M over a principal ideal domain R is free if and only if M is projective.*

§5.2

Theorem 5.4. *A finitely generated torsion-free module M over a principal ideal domain R is free.*

Proof. Let M be generated by n elements. By Corollary 6.2, every submodule of M may be generated by m elements with $m \leq n$. We proceed by induction on n .

If $n = 1$, then $M \cong R/\text{Ann}(M)$ is torsion-free, whence $\text{Ann}(M) = 0$ and $M \cong R$ is free.

Suppose the theorem is true for $n - 1$ and let M be generated by n elements. Let N be a maximal submodule of M ; then N may be generated by m elements with $m \leq n$. Since M/N is

cyclic, there exists an epimorphism

$$f : R \rightarrow M/N$$

with $\text{Ker } f = \text{Ann}(M/N)$. If $\text{Ann}(M/N) \neq 0$, then M/N has torsion, which contradicts the fact that M is torsion-free. Therefore, $\text{Ann}(M/N) = 0$ and $M/N \cong R$ is free.

By the induction hypothesis, N is free since it is generated by at most $n - 1$ elements. Consequently, the exact sequence

$$0 \rightarrow N \xrightarrow{\subseteq} M \rightarrow M/N \rightarrow 0$$

is split exact and $M \cong N \oplus M/N$ by Theorem 2.6 and Proposition 2.7. Therefore, M is free. $\square$

Theorem 5.5. *If M is a finitely generated module over R , then*

$$M = \text{Tor}(M) \oplus F$$

where F is a free left R -module of finite rank and $F \cong M/\text{Tor}(M)$.

Proof. The quotient module M/M_t is torsion-free since for each $r \neq 0$,

$$r(a + M_t) = M_t \Rightarrow ra \in M_t \Rightarrow r_1(ra) = 0 \text{ for some } r_1 \neq 0 \Rightarrow a \in M_t$$

Furthermore, M/M_t is finitely generated since M is. Therefore, M/M_t is free of finite rank by Theorem 6.5. Consequently, the exact sequence

$$0 \rightarrow M_t \xrightarrow{\subseteq} M \rightarrow M/M_t \rightarrow 0$$

is split exact and $M \cong M_t \oplus (M/M_t)$ by 2.8 and ??.

Under the isomorphism $M_t \oplus M/M_t \cong M$ of Theorem 3.4 the image of M_t is M_t and the image of M/M_t is a submodule F of M , which is necessarily free of finite rank. It follows that M is the internal direct sum $M = M_t \oplus F$ (see Theorem 1.15). $\square$

§5.3 Torsion module decomposition

Theorem 5.6. *Let M be a torsion module over a principal ideal domain R and for each prime $p \in R$ let $M_p = \{m \in M \mid m \text{ has order a power of } p\}$.*

1. M_p is a submodule of M for each prime $p \in R$.
2. $M = \bigoplus M_p$ where the sum is over all primes $p \in R$. If A is finitely generated, only finitely many of the M_p are nonzero.

Proof. Let $0 \neq a \in M$ with $\text{Ann}(a) = (r)$. By Theorem III.3.7 $r = p_1^{n_1} \cdots p_k^{n_k}$ with p_i distinct primes in R and each $n_i > 0$. For each i , let $r_i = p_1^{n_1} \cdots p_{i-1}^{n_{i-1}} p_{i+1}^{n_{i+1}} \cdots p_k^{n_k}$. Then $r_1, \dots, r_k$ are relatively prime and there exist $s_1, \dots, s_k \in R$ such that $s_1 r_1 + \cdots + s_k r_k = 1_R$ by 2.2.

Consequently, $a = 1_R a = s_1 r_1 a + \cdots + s_k r_k a$, and we have proved that the submodules M_p generate the module M .

Let $p \in R$ be prime and let M_1 be the submodule of M generated by all M_q with $q \neq p$. Suppose $a \in M(p) \cap A_1$. Then $p^m a = 0$ for some $m \geq 0$ and $a = a_1 + \cdots + a_t$ with $a_i \in A(q_i)$ for some primes $q_1, \dots, q_t$ all distinct from p . Since $a_i \in A(q_i)$, there are integers m_i such that $q_i^{m_i} a_i = 0$, whence $(q_1^{m_1} \cdots q_t^{m_t}) a = 0$. If $d = q_1^{m_1} \cdots q_t^{m_t}$, then p^m and d are relatively prime and $rp^m + sd = 1_R$ for some $r, s \in R$. Consequently, $a = 1_R a = rp^m a + sda = 0$. Therefore, $A(p) \cap A_1 = 0$ and $A = \sum A(p)$ by Theorem 1.15. The last statement of the Theorem is a consequence of the easily verified fact that a direct sum of modules with infinitely many nonzero summands cannot be finitely generated. For each generator has only finitely many nonzero coordinates. $\square$

Lemma 5.7. *Let M be a unitary module over a principal ideal domain R such that $p^n M = 0$ and $p^{n-1} M \neq 0$ for some prime $p \in R$ and positive integer n . Let a be an element of M of order p^n .*

1. *If $M \neq Ra$, then there exists a nonzero $b \in M$ such that $Ra \cap Rb = 0$.*
2. *There is a submodule N of M such that $M = Ra \oplus N$.*

Theorem 5.8. *Let M be a finitely generated unitary module over a principal ideal domain R such that every element of M has order a power of some prime $p \in R$. Then M is a direct sum of cyclic R -modules of orders $p^{n_1}, \dots, p^{n_k}$ respectively, where $n_1 \geq n_2 \geq \cdots \geq n_k \geq 1$.*

§5.4

Theorem 5.9. *Let M be a finitely generated unitary module over a principal ideal domain R .*

1. *M is the direct sum of a free submodule F of finite rank and a finite number of cyclic torsion modules. The cyclic torsion summands (if any) are of orders $r_1, \dots, r_t$, where $r_1, \dots, r_t$ are (not necessarily distinct) nonzero nonunit elements of R such that $r_1 \mid r_2 \mid \cdots \mid r_t$. The rank of F and the list of ideals $(r_1), \dots, (r_t)$ are uniquely determined by M .*
2. *M is the direct sum of a free submodule E of finite rank and a finite number of cyclic torsion modules. The cyclic torsion summands (if any) are of orders $p_1^{s_1}, \dots, p_k^{s_k}$, where $p_1, \dots, p_k$ are (not necessarily distinct) primes in R and $s_1, \dots, s_k$ are (not necessarily distinct) positive integers. The rank of E and the list of ideals $(p_1^{s_1}), \dots, (p_k^{s_k})$ are uniquely determined by M .*

The elements $r_1, \dots, r_t$ are called the **invariant factors** of the module M just as in the special case of abelian groups. Similarly $p_1^{s_1}, \dots, p_k^{s_k}$ are called the **elementary divisors** of M .

Part IV

Field and Galois Theory

Chapter IX

Field Theory

§1 Field Extension

§1.1 Basic Definition

Definition 1.1. Let K be a fixed field. We define the category of field extensions $\mathbf{Field}/K$ as follows:

- An object, called **extension**, is field L together with a fixed embedding $i_L : K \hookrightarrow L$. We typically suppress i_L and simply write L/K .
- A morphism from F/K to L/K is a field homomorphism $\phi : F \rightarrow L$ such that the following diagram commutes:

$$\begin{array}{ccc} K & \xrightarrow{i_F} & F \\ & \searrow i_L & \downarrow \phi \\ & & L \end{array}$$

simply denoted as $\phi|_K = \text{id}_K$. The morphism ϕ is called a **K -homomorphism (embedding)** and field F is called an **intermediate field** of L/K . The all morphisms from L_1/K to L_2/K is denoted as $\text{Hom}_K(L_1, L_2)$.

Remark. Let $L_1 = \mathbb{Q} \times \{1\}$ and $L_2 = \mathbb{Q} \times \{2\}$, then $L_1/\mathbb{Q}$ and $L_2/\mathbb{Q}$ are both field extensions but $L_1 \cap L_2 = \emptyset$.

§1.2 Composition fields

Definition 1.2. Let $\mathbf{Field}/K$ be the category of field extensions of K and L_1/K , L_2/K be two extension. The **composition** of L_1/K and L_2/K is an extension Ω/K such that

- L_i is an intermediate field of Ω/K (Ω is a **overfield** of both L_1 and L_2).
- If extension Ω'/K satisfies the above condition, then Ω is an intermediate field of Ω'/K .

Theorem 1.3. *The composition of two field extensions L_1/K and L_2/K in $\mathbf{Field}/K$ exists and uniquely up to K -isomorphism (is isomorphic in $\mathbf{Field}/K$).*

Proof. Let $A = L_1 \otimes_K L_2$ be the pushout (tensor product of K -algebra L_i) of $L_1 \leftarrow K \rightarrow L_2$ in the category $\mathbf{CRing}$ and there is a prime ideal $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is an integral domain with fractions field $\Omega = \text{Frac}((L_1 \otimes_K L_2)/\mathfrak{p})$. The following diagram commutes:

$$\begin{array}{ccc}
 K & \longrightarrow & L_2 \\
 \downarrow & & \downarrow \\
 L_1 & \longrightarrow & A \\
 & \searrow & \downarrow \\
 & & \Omega
 \end{array}$$

thus L_i can be imbedded into Ω by

$$L_i \rightarrow A \rightarrow A/\mathfrak{p} \rightarrow \Omega$$

respectively, Ω is an overfield of both L_1 and L_2 .

If there is another overfield Ω' of both L_1 and L_2 , then by the universal property of pushout, Ω can be imbedded into Ω' . $\square$

By theorem 1.3, when studying a collection of extensions L_i/K , we can have an overfield Ω and a common embedding ι such that $\iota(K) \subset \iota(L_i) \subset \Omega$ for each i .

§2 Extension tower

Definition 2.1. *In the category of field extensions $\mathbf{Field}/K$,*

1. The **degree** of a field extension L/K , denoted $[L : K]$, is the dimension of L as a vector space over K .
2. L is said to be a **finite dimensional extension** or **infinite dimensional extension** of K according as $[L : K]$ is finite or infinite.

It follows that $[L : K] = [L : F][F : K]$. Furthermore $[L : K]$ is finite if and only if $[L : F]$ and $[F : K]$ are finite.

Theorem 2.2. *Let field extension $K \subset L, M$. The following statements hold:*

1. If $[LM : K]$ is finite, then $[L : K]$ and $[M : K]$ divide $[LM : K]$ and

$$\begin{aligned}
 [LM : K] &= [LM : L][L : K] \\
 &= [M : L \cap M][L : K] \\
 &\leq [M : K] \leq [L : K]
 \end{aligned}$$

Corollary 2.3. *Let L and M be intermediate fields in the extension F/K .*

1. *$[LM : K]$ is finite if and only if $[L : K]$ and $[M : K]$ are finite.*
2. *If $[L : K]$ and $[M : K]$ are finite and relatively prime, then*

$$[LM : K] = [L : K][M : K]$$

3. *If L and M are algebraic over K , then so is LM .*
4. *Assume that $[LM : K] = [L : K][M : K]$, then $L \cap M = K$.*

§3 Generation

Definition 3.1. *Let L/K be a field extension and a subset $X \subset L$*

1. *the **subfield generated by X over K** is the intersection of all subfields of L that contain $X \cup K$, denoted by $K(X)$.*
2. *If $X = \{u_1, \dots, u_n\}$, then the subfield $F(X)$ of K is denoted $K(u_1, \dots, u_n)$. The field $K(u_1, \dots, u_n)$ is said to be a **finitely generated extension of K** . If $X = \{u\}$, then $F(u)$ is said to be a **simple extension of F** and u is said **primitive element**.*
3. *If F is a field and $X \subset F$, then the subring generated by X is the intersection of all subrings of F that contain X . If F is an extension field of K and $X \subset F$, then the subring generated by $K \cup X$ is called the subring generated by X over K and is denoted $K[X]$.*
4. *If $X = \{u_1, \dots, u_n\}$, then the subring $K[X]$ of F is denoted $K[u_1, \dots, u_n]$.*

Theorem 3.2. *Let L/K be a field extension, $u, u_i \in L$, and $X \subset L$, then*

1. *The subring $K[X]$ consists of all elements of the form $h(u_1, \dots, u_n)$, where each $u_i \in X$, n is a positive integer, and $h \in K[x_1, \dots, x_n]$.*
2. *The subfield $K(X)$ consists of all elements of the form*

$$f(u_1, \dots, u_n) / g(u_1, \dots, u_n) = f(u_1, \dots, u_n) g(u_1, \dots, u_n)^{-1}$$

where $n \in \mathbb{Z}_{>0}$, $f, g \in K[x_1, \dots, x_n]$, $u_1, \dots, u_n \in X$ and $g(u_1, \dots, u_n) \neq 0$.

3. *For each $v \in K(X)$ (resp. $K[X]$) there is a finite subset X' of X such that $v \in K(X')$ (resp. $K[X']$). Furthermore, we have that*

$$K(X) = \bigcup_{\#X' < \infty} K(X'), \quad K[X] = \bigcup_{\#X' < \infty} K[X']$$

Corollary 3.3. *For any $u_1, \dots, u_n \in F$ and any permutation $\sigma \in S_n$.*

1. $K(u_1, \dots, u_n) = K(u_{\sigma(1)}, \dots, u_{\sigma(n)})$.
2. $K(u_1, \dots, u_{n-1})(u_n) = K(u_1, \dots, u_n)$.
3. $K[u_1, \dots, u_n] = K[u_{\sigma(1)}, \dots, u_{\sigma(n)}]$.
4. $K[u_1, \dots, u_{n-1}][u_n] = K[u_1, \dots, u_n]$.

§3.1 Finitely Generated Extensions

Definition 3.4. *Let F be an extension field of K .*

1. *An element u of F is said to be **algebraic over K** provided that u is a root of some nonzero polynomial $f \in K[x]$. F is called an **algebraic extension** of K if every element of F is algebraic over K .*
2. *If u is not a root of any nonzero $f \in K[x]$, u is said to be **transcendental over K** . F is called a **transcendental extension** if at least one element of F is transcendental over K .*
3. *Let $u_1, \dots, u_n$ be element of F , then u_i are **algebraically independent** provided that there is no nonzero polynomial $f \in K[x_1, \dots, x_n]$ such that $f(u_1, \dots, u_n) = 0$.*

Remark. *It follows that each u_i is transcendental.*

§3.2 Simple Extension

Definition 3.5. *Let L/K be an extension field and $u \in L$ algebraic over K . The monic minimal polynomial $m_u(X)$ is called the **irreducible (or minimal or minimum) polynomial** of u .*

Theorem 3.6. *If L is an extension field of K and $u \in L$ is algebraic over K , then*

1. $K(u) = K[u]$
2. $K(u) \cong K[x]/(m_u)$
3. $\{1, u, u^2, \dots, u^{n-1}\}$ is a K -basis of $K(u)$, where $n = \deg m_u$

Corollary 3.7 (Adjoining a root). *Let K be a field and $f \in K[x]$ be a irreducible polynomial. Then there exists a simple extension field $L = K(u)$ such that:*

1. $u \in L$ is a root of f
2. $[L : K] = n$, where $n = \deg f$;
3. L is unique up to an K -isomorphism

Theorem 3.8. *If $u \in F$ is transcendental over K , then*

1. K -isomorphism of fields $K(u) \cong K(x)$.
2. K -algebra isomorphism $K[u] \cong K[x]$.

Theorem 3.9 (Uniqueness of simple extension). *Let $\sigma : K_1 \rightarrow K_2$ be an isomorphism of fields, u an element of some extension field of K_1 and v an element of some extension field of K_2 . Assume either*

- u is transcendental over K_1 and v is transcendental over K_2 ; or
- u is a root of an minimal polynomial $f \in K_1[x]$ and v is a root of $\sigma f \in K_2[x]$.

Then σ extends to an isomorphism of fields $K_1(u) \cong K_2(v)$ which maps u onto v .

§3.3 n -extension

Theorem 3.10. *If $u_1, \dots, u_n \in F$ then the field $K(u_1, \dots, u_n)$ is isomorphic to the quotient field of the ring $K[u_1, \dots, u_n]$.*

Proof. By first homomorphism theorem, we have

$$K[X_1, \dots, X_n]/I \cong K(u_1, \dots, u_n)$$

where I denotes the ideal $\{f(u_1, \dots, u_n) = 0 : f \in K[X_1, \dots, X_n]\}$ □

Theorem 3.11. *Let F be an extension of K .*

1. *If each u_i is algebraic over K , then $K(u_1, \dots, u_n) = K[u_1, \dots, u_n]$.*
2. *If v_i are algebraically independent then $K(v_1, \dots, v_n) \cong K(x_1, \dots, x_n)$.*

§4 Algebraic Extension

In this section, we always assume that all extension L_i/K encountered in a problem are contained in a fixed overfield Ω and $K \subset L_i \subset \Omega$ as before.

Theorem 4.1. *Let F be an extension of K if and only if for every intermediate field E every monomorphism $\sigma : E \rightarrow E$ which is the identity on K is in fact an automorphism of E .*

Proof. Suppose F/K is an algebraic extension and E be an intermediate field, thus E/K is also an algebraic extension. For every monomorphism $\sigma : E \rightarrow E$ which is the identity on K , if $\sigma(E) \neq E$, there is an element $\alpha \in E - \sigma(E)$.

Let f be the minimal polynomial of α in $K[x]$, and $\alpha_1, \alpha_2, \dots, \alpha_s$ be all roots of f in E . We have

$$0 = \sigma f(\alpha_k) = \sigma(f)(\sigma \alpha_k) = f(\sigma \alpha_k)$$

It follows from that f is injective, that $\sigma \alpha_k$ is a permutation of $\{\alpha_k\}$. It contradicts the assumption. □

Theorem 4.2. *Let K be a field.*

1. *If L/K is finite, then L is algebraic over K .*
2. *If $\{L_i/K\}_{i \in I}$ be algebraic extensions, then the composition field L is also algebraic over K .*
3. *If L/F and F/K are algebraic extensions, then so is L/K .*

Theorem 4.3. *Let L/K be a field extension and E the set of all elements of L which are algebraic over K . Then E is a subfield of L called **maximal algebraic extension of K/L** .*

Proof. If $u, v \in E$, then $K(u, v)$ is an algebraic extension field of K . Therefore, since $u - v$ and $uv^{-1}(v \neq 0)$ are in $K(u, v)$, $u - v$ and $uv^{-1} \in E$. This implies that E is a field. $\square$

Theorem 4.4. *The following conditions on a field K are equivalent.*

1. *Every nonconstant polynomial $f \in K[x]$ has a root in K .*
2. *every nonconstant polynomial $f \in K[x]$ splits over K .*
3. *every minimal polynomial in $K[x]$ has degree one.*
4. *there is no algebraic extension field of K except K itself.*
5. *there exists a subfield K of F such that F is algebraic over K and every polynomial in $K[x]$ splits in $F[x]$.*

*A field K that satisfies the equivalent conditions is said to be **algebraically closed**.*

Definition 4.5. *Let L/K be a field extension, then the following conditions are equivalent.*

1. *L is algebraic over K and L is algebraically closed.*
2. *L is a splitting field of $K[x]$ over K .*

*The field L that satisfies the equivalent conditions is called an **algebraic closure of K** .*

Theorem 4.6. *Let K be a field. Then the algebraic closure of K exists and is unique up to an isomorphism in **Field**/ K .*

Proof. Let $\mathcal{F} := \{L : L/K \text{ is finite}\}$ be a collection of extensions (assume that all L is contained in a fixed overfield by theorem 1.3) with partial order defined by $L_1 \leq L_2 \Leftrightarrow L_1 \subset L_2$. Then $\mathcal{F}$ is a directed system in **Field**/ K . Thus the direct limit

$$\varinjlim L = \bigcup_{L \in \mathcal{F}} L$$

exists (be a field containing all L and K) and is algebraic over K by theorem 4.2 and algebraically closed by corollary 3.7. $\square$

Corollary 4.7. *If L_i/K is algebraic extensions for each i , then there is a algebraic closure $\bar{K}$ of K containing all L_i .*

§4.1 Splitting Fields

Definition 4.8. Let K be a field and $f \in K[x]$ a polynomial of positive degree. An extension field $L \supset K$ is said to be a **splitting field** over K of the polynomial f if

- (i) f splits in $L[x]$
- (ii) $L = K(u_1, \dots, u_n)$ where $u_1, \dots, u_n$ are the roots of f in L .

Let $\mathcal{S}$ be a set of polynomials of positive degree in $K[x]$. An extension field L of K is said to be a **splitting field over K of the set $\mathcal{S}$** if

- (i) every polynomial in $\mathcal{S}$ splits in $L[x]$.
- (ii) L is generated over K by the roots of all the polynomials in $\mathcal{S}$.

Remark. L is a splitting field over K of a finite set $\{f_1, \dots, f_n\} \subset K[x]$ if and only if L is a splitting field over K of the single polynomial $f = f_1 f_2 \cdots f_n$.

If F is a splitting field of $\mathcal{S}$ over K , then F is also a splitting field over K of the set $\mathcal{T}$ of all irreducible factors of polynomials in $\mathcal{S}$.

Theorem 4.9 (Existence of splitting field). If K is a field and $f \in K[x]$ has degree $n \geq 1$, then there exists a splitting field L of f with $[L : K] \leq n!$

Theorem 4.10. Let $\sigma : K \rightarrow K'$ be an isomorphism of fields, $\mathcal{S} = \{f_i\} \subset K[x]$, and $\mathcal{S}' = \{f'_i = \sigma f_i\}$. If L is a splitting field of $\mathcal{S}$ over K and L' is a splitting field of $\mathcal{S}'$ over K' , then σ is extendible to an isomorphism $L \cong L'$.

Proof. Step 1. Suppose first that $\mathcal{S}$ consists of a single polynomial $f \in K[x]$ and proceed by induction on $n = [L : K]$. If $n = 1$, then $L = K$ and f splits over K . This implies that σf splits over K' and hence that $L' = K'$. Thus σ itself is the desired isomorphism $L = K \xrightarrow{\sigma} K' = L'$.

If $[L : K] = > 1$, then f must have an irreducible factor g of degree greater than 1. Let u be a root of g in L . Then verify that σg is irreducible in $K'[x]$. If v is a root of σg in L' , then σ extends to an isomorphism $\tau : K(u) \cong K'(v)$ with $\tau(u) = v$. Since $[K(u) : K] = \deg g > 1$, we must have $[L : K(u)] < n$. Since L is a splitting field of f over $K(u)$ and L' is a splitting field of σf over $K'(v)$, the induction hypothesis implies that τ extends to an isomorphism $L \cong L'$.

Step 2. If $\mathcal{S}$ is arbitrary, let $\mathcal{S}$ consist of all triples (E, E', τ) , where E is an intermediate field of F and K , E' is an intermediate field of F' and K' , and $\tau : E \rightarrow E'$ is an isomorphism that extends σ .

Define $(E_1, E'_1, \tau_1) \leq (E_2, E'_2, \tau_2)$ if $E_1 \subset E_2$, $E'_1 \subset E'_2$ and $\tau_2 \mid \tau_1$. Verify that $\mathcal{S}$ is a nonempty partially ordered set in which every chain has an upper bound in $\mathcal{S}$. By Zorn's Lemma there is a maximal element (E_0, E'_0, τ_0) of $\mathcal{S}$. We claim that $E_0 = F$ and $E'_0 = F'$, so that $\tau_0 : F \cong F'$ is the desired extension of σ by the maximality and Step 1. $\square$

Corollary 4.11 (Uniqueness of splitting field). Let K be a field and $\mathcal{S} \subset K[x]$. Then any two splitting fields of $\mathcal{S}$ are K -isomorphic. In particular, any two algebraic closures of K are K -isomorphic.

§4.2 Normal Extension

Definition 4.12. An algebraic extension field L/K is said **normal** if every irreducible polynomial in $K[x]$ that has a root in L actually splits in $L[x]$.

Theorem 4.13. Let L/K be an algebraic extension and $\overline{K}$ be an algebraic closure of K containing L , then the following statements are equivalent.

1. L is normal over K .
2. L is a splitting field over K of some set $S \subset K[x]$.
3. $\sigma(L) = L$ for all $\sigma \in \text{Hom}_K(L, \overline{K})$, that is $\text{Hom}_K(L, \overline{K}) = \text{Aut}_K L$

Proof. (1) $\Rightarrow$ (2). Let $\{u_i \mid i \in I\}$ be a basis of vector space L over K and for each $i \in I$ let $f_i \in K[x]$ be the minimal polynomial of u_i . Since L/K is normal, each f_i splits in $L[x]$. Therefore L is a splitting field over K of $S = \{f_i \mid i \in I\}$.

(2) $\Rightarrow$ (3). Let u be a root of some polynomial in S . Since L is a splitting field of S over K , we have $u \in L$. For any $\sigma \in \text{Hom}_K(L, \overline{K})$, $\sigma(u)$ is also a root of the same polynomial. Thus $\sigma(u) \in L$. Since L is generated over K by the roots of all polynomials in S , we have σ maps each generator of L into L . It follows that $\sigma(L) \subset L$. Since σ is injective, we have $\sigma(L) = L$.

(3) $\Rightarrow$ (1). Let $f \in K[x]$ be an irreducible polynomial with a root $u \in L$. If v is any root of f in $\overline{K}$, then there is a $\sigma \in \text{Hom}_K(K(u), \overline{K})$ such that $\sigma(u) = v$. Since L/K is algebraic, by extending σ , we have $\sigma \in \text{Hom}_K(L, \overline{K})$. By assumption, we have $\sigma(L) = L$, thus $v = \sigma(u) \in L$. Therefore, $\{\sigma(u)\}$ runs over all roots of f and f splits in $L[x]$. $\square$

Proposition 4.14. Let k be a field. The following statements hold:

1. If $F \supset L \supset k$ and F is normal over k , then F is normal over L .
2. If L_1, L_2 are normal over k , then $L_1 L_2$ is normal over k , and so is $L_1 \cap L_2$.

Definition 4.15. Let L/K be an algebraic extension, the **normal closure** of L over K is the smallest normal extension of K containing L , that is, an extension field $\tilde{L}$ of L such that

- (i) $\tilde{L}$ is normal over K .
- (ii) no proper subfield of $\tilde{L}$ containing L is normal over K .

Remark. The normal closure of L over K exists ($\overline{K}$ is normal over K and contains L) and is unique up to a K -isomorphism.

$$\bigcap_{\substack{L \subset F \subset \overline{K} \\ F \text{ normal over } K}} F$$

§4.3 Separable Extension

Definition 4.16. Let L/K be an algebraic extension.

1. The polynomial $f \in K[x]$ is said to be **separable** if every root of f is a simple root in some splitting field of f over K .
2. Let $u \in L$ be algebraic over K , then u is said to be **separable** over K provided its minimal polynomial is separable. It is equivalent that $\gcd(m_u, m'_u) = 1$.
3. If every element of F is separable over K , then F is said to be a **separable extension** of K . (thus algebraic extension)

Remark. Thus separable polynomial has distinct roots in its splitting field. If $\text{char } K = 0$, every irreducible polynomial in $K[x]$ is separable and every algebraic extension L/K is separable.

Proposition 4.17. Let L/K be an algebraic extension. If L is generated by a set of separable elements over K , then L is a separable extension of K .

Definition 4.18. Let L/K be an algebraic extension, the **separable closure** of L over K is the largest separable extension of K contained in L , that is, an extension field L_{sep} of K such that

- (i) L_{sep} is separable over K .
- (ii) any proper extension field of L_{sep} contained in L is not separable over K .

The separable degree of L over K is defined as $[L_{\text{sep}} : K]$, denoted by $[L : K]_s$.

Remark. That is,

$$L_{\text{sep}} = \{u \in L : u \text{ is separable over } K\}$$

§4.4 Purely Inseparable Extension ($\text{char } p$)

Definition 4.19. Let L/K be an algebraic extension (of characteristic p) and an element $\alpha \in L$ with minimal polynomial $m_\alpha \in K[x]$.

1. The **separable degree** of a polynomial $f \in K[x]$ is defined as the number of distinct roots of f in its splitting field over K , denoted by $\deg_s(f)$. And the **separable part** of f is defined as $f_{\text{sep}}(x) := \prod (x - \alpha_i)$
2. the **inseparable degree** of f is defined as $\deg_i(f) := \deg f / \deg_s(f)$.
3. The **separable degree of α over K** is

$$\deg_s \alpha := \text{number of distinct roots of } m_\alpha \text{ in its splitting field over } K$$

4. A element $u \in L$ is **purely inseparable** over K if its minimal polynomial f in $K[x]$ factors in $L[x]$ as $f = (x - u)^m$ (or equivalently $X^{p^n} - a$ for some $a \in K$ and $n \in \mathbb{Z}_{\geq 1}$).

5. L is a **purely inseparable extension** of K if every element of L is purely inseparable over K .

Theorem 4.20. *Let L/K be an algebraic extension (of characteristic p), then the following statements are equivalent:*

1. L is purely inseparable over K ;
2. the minimal polynomial of any $u \in L$ is of the form $x^{p^n} - a \in K[x]$;
3. if $u \in L$, then $u^{p^n} \in K$ for some $n \geq 0$;
4. the only elements of L which are separable over K are the elements of K itself;
5. L is generated over K by a set of purely inseparable elements.

$$\text{Hom}_K(L, L)$$

Theorem 4.21 (Decomposition of algebraic extension). *Let L/K be an algebraic extension, then there exists a unique intermediate field M (actually L_{sep}) such that L_{sep}/K is separable and L/L_{sep} is purely inseparable.*

Proof. If $\text{char } K = 0$, then let $M = L$. In the case $\text{char } K = p \neq 0$, let $u \in L$ and m_u be the minimal polynomial of u over K . We can write $m_u(x) = f(x^{p^n})$ for some irreducible separable polynomial $f \in K[x]$ and integer $n \geq 0$. Thus f is the minimal polynomial of u^{p^n} over K and $u^{p^n} \in L_{\text{sep}}$ is separable over K . Then u is purely inseparable over L_{sep} since its minimal polynomial in $L_{\text{sep}}[x]$ divides $X^{p^n} - u^{p^n} = (X - u)^{p^n}$. $\square$

§4.5 Separable degree

Theorem 4.22 (Primitive element theorem). *Let L/K be a finite extension, then the following statements are equivalent.*

1. there exists an element $u \in L$ such that $L = K(u)$.
2. there exists only a finite number of intermediate field such that $K \subset F \subset L$

Epecially, if L/K is finite separable, then $L = K(u)$ for some $u \in L$.

Corollary 4.23. *Let L/K be an algebraic extension and $\overline{K}$ be an algebraic closure of K containing L , then the **separable degree***

$$[L : K]_s := [L_{\text{sep}} : K] = \# \text{Hom}_K(L, \overline{K})$$

(if finite)

Definition 4.24. *Let L/K be an algebraic extension*

§4.6

Theorem 4.25. *Let L/K be an algebraic extension, then the following statements are equivalent.*

1. L/K is Galois.
2. L is a splitting field over K of and L/K is separable.
3. L is a splitting field over K of a set S of separable polynomials in $K[x]$.

Proof. (1) $\Rightarrow$ (2). Suppose $u \in F$ has minimal polynomial $f \in K[x]$, then f splits in $F[x]$ into a product of distinct linear factors. Hence u is separable over K . Let $\{v_i \mid i \in I\}$ be a basis of vector space F over K and for each $i \in I$ let $f_i \in K[x]$ be the minimal polynomial of v_i . The preceding remarks show that each f_i is separable and splits in $F[x]$. Therefore F is a splitting field over K of $S = \{f_i \mid i \in I\}$.

(2) $\Rightarrow$ (3) Let set T consists of all irreducible monic factor of polynomial in S . Let $f \in T$, f must be the minimal polynomial of some $u \in F$. Since F is separable over K , f is necessarily separable. It follows that F is a splitting field over K of the set T of separable polynomials consisting of all monic irreducible factors in $K[x]$.

(3) $\Rightarrow$ (1) F is algebraic over K since any splitting field over K is an algebraic extension. If $u \in F - K$, then $u \in K(v_1, \dots, v_n)$ with each v_i a root of some $f_i \in T$. □

Chapter X

Galois Theory

§1 Basic Definition

Definition 1.1. Let E and F be extension fields of a field K .

1. A nonzero map $\sigma : E \rightarrow F$ which is both a field homomorphism and a K -module homomorphism is called a **K -homomorphism**.
2. Similarly if a field automorphism $\sigma \in \text{Aut } F$ is a K -homomorphism, then σ is called a **K -automorphism** of F .
3. The group of all K -automorphisms of F is called the **Galois group of F over K** and is denoted $\text{Aut}_K F$ or $\text{Gal}(F/K)$.

Definition 1.2. Let L/K be an extension.

1. If $H < \text{Aut}_K F$, then

$$H' = L^H := \{v \in F : \sigma(v) = v \text{ for all } \sigma \in H\}$$

is an intermediate field of the extension called the **fixed field of H in F** .

2. If E an intermediate field, then

$$E' = \text{Aut}_E F = \{\sigma \in \text{Aut}_K F : \sigma(u) = u \text{ for all } u \in E\}$$

is a subgroup of $\text{Aut}_K F$.

Definition 1.3. Let L/K be an algebraic extension such that $K = L^{\text{Aut}_K(L)}$. Then L is said to be a **Galois extension of K** or to be **Galois over K** .

Let X be an intermediate field or subgroup of the Galois group. X will be called **closed** provided $X = X''$.

§2

Proposition 2.1. *Let L/K be an algebraic extension and $\overline{K}$ be an algebraic closure of K containing L , then*

$$[L : K]_{sep} = \# \text{Hom}_K (L, \overline{K})$$

if finite. Thus we have

$$|\text{Aut}_K L| \cdot t = [L : K]_s$$

where $t = \# \{ \sigma(L) : \sigma \in \text{Hom}_K (L, \overline{K}) \}$ is the number of distinct K -embeddings image of L , is also the number of distinct conjugates of L in $\overline{K}$.

Corollary 2.2. *Let L/K be an algebraic extension and intermediate field $E \subset F$, then*

$$[\text{Aut}_E L : \text{Aut}_F L] \cdot \frac{t_E}{t_F} = [F : E]_{sep}$$

Corollary 2.3. *If L/K is a finite extension, then*

$$|\text{Aut}_K L| \leq [L : K]$$

The equality holds if and only if L/K is finite separable and normal.

Proposition 2.4. *Let L be a field, $H < \text{Aut}(L)$ and if L/L^H is algebraic, then L/L^H is Galois.*

Proof. In any case H is a subgroup of $\text{Aut}_{L^H} L$. If $u \in L - L^H$, then there must be a $\sigma \in H$ such that $\sigma(u) \neq u$. Therefore, the fixed field of $\text{Aut}_{L^H} L$ is L^H , whence L is Galois over L^H . $\square$

§3 Fundamental Theorem

Proposition 3.1. *Let L/K be an extension field.*

1. *Suppose that $E \subset F$ are intermediate fields, then σ_1 and $\sigma_2 \in \text{Aut}_E L$ are in the same left coset of F' if and only if*

$$\sigma_1|_F = \sigma_2|_F$$

thus $[\text{Aut}_E F : \text{Aut}_F F] = \# \{ \sigma|_F : \sigma \in \text{Aut}_E L \}$.

2. *Suppose that H, J are subgroups of $\text{Aut}_K L$ with $H < J$ and $\tau_1, \tau_2 \in J$ are in the same left coset of H , then*

$$\tau_1|_{L^H} = \tau_2|_{L^H}$$

thus $[J : H] \geq \# \{ \tau|_{L^H} : \tau \in J \}$.

Lemma 3.2. *Let L/K be an extension field and $E \subset F$ are intermediate fields, then σ_1 and $\sigma_2 \in \text{Aut}_E L$ are in the same left coset of F' if and only if*

$$\sigma_1|_F = \sigma_2|_F$$

thus $[\text{Aut}_E F : \text{Aut}_F F] = \#\{\sigma|_F : \sigma \in \text{Aut}_E L\}$.

Theorem 3.3. *Let L/K be an algebraic extension and $E \subset F$ intermediate fields with $[F : E] < \infty$.*

$$[\text{Aut}_E L : \text{Aut}_F L] \leq [F : E]$$

Lemma 3.4. *Suppose that H, J are subgroups of $\text{Aut}_K L$ with $H < J$ and $\tau_1, \tau_2 \in J$ are in the same left coset of H , then*

$$\tau_1|_{L^H} = \tau_2|_{L^H}$$

thus $[J : H] \geq \#\{\tau|_{L^H} : \tau \in J\}$.

Theorem 3.5. *Let χ_i be distinct characters of a group G with degree n_i , then*

Theorem 3.6. *Let L be a field, G be a finite subgroup of $\text{Aut}(L)$ and $K = L^G$, then*

$$[L : K] = |G|$$

Proof.

□

Corollary 3.7. *Let L/K be an algebraic extension and H, J be subgroups of $\text{Aut}_K L$ with $[J : H] < \infty$, then*

$$[L^H : L^J] \leq [J : H]$$

Lemma 3.8. *Let F/K and X, Y be two intermediate fields or two subgroups of the Galois group $\text{Aut}_K F$. Then:*

1. $X \subset Y \Rightarrow Y' \subset X'$
2. $X' = X'''$

Lemma 3.9. *Let L/K , then there is a one-to-one correspondence between the closed intermediate fields and the closed subgroups, given by*

$$E \mapsto E'$$

$$H \mapsto H'$$

Corollary 3.10. *Let F be an extension field of K , L and M intermediate fields with $L \subset M$, and H, J subgroups of $\text{Aut}_K F$ with $H < J$.*

1. *If L is closed and $[M : L]$ finite, then M is closed and $[L' : M'] = [M : L]$*
2. *If H is closed and $[J : H]$ finite, then J is closed and $[H' : J'] = [J : H]$*
3. *If F is a finite dimensional Galois extension of K , then all intermediate fields and all subgroups of the Galois group are closed.*

Proof. (2) Applying successively the facts that $J \subset J''$ and $H = H''$ and Lemmas 2.8 and 2.9 yields

$$[J : H] \leq [J'' : H] = [J'' : H''] \leq [H' : J'] \leq [J : H];$$

this implies that $J = J''$ and $[H' : J'] = [J : H]$. (1) is proved similarly. $\square$

§3.1 Stable Intermediate Fields

Definition 3.11. Let E be an intermediate field of the extension L/K

1. The intermediate field E is said to be **stable relative to K and L** if every K -automorphism $\sigma \in \text{Aut}_K F$ maps E into itself.

Remark. If E is stable and $\sigma^{-1} \in \text{Aut}_K F$ is the inverse automorphism, thus σ^{-1} also maps E into itself. This implies that $\sigma|_E$ is in fact a K -automorphism of E (that is, $\sigma|_E \in \text{Aut}_K E$) with inverse $\sigma^{-1}|_E$.

2. A K -automorphism $\tau \in \text{Aut}_K E$ is said to be **extendible** to F if there exists $\sigma \in \text{Aut}_K F$ such that $\sigma|_E = \tau$.

It is easy to see that the extendible K -automorphisms form a subgroup of $\text{Aut}_K E$.

Theorem 3.12. Let L/K be an extension and E be an intermediate field.

1. If E is a stable intermediate field, then

$$E' = \text{Aut}_E L \triangleleft \text{Aut}_K L$$

and the quotient group

$$G/E' \cong \{\sigma \in \text{Aut}_K E : \sigma \text{ is extendible to } F\}$$

2. If $H \triangleleft \text{Aut}_K F$, then H' is a stable intermediate field.

Proof. (1) If $u \in E$ and $\sigma \in \text{Aut}_K F$, then $\sigma(u) \in E$ by stability and hence $\tau\sigma(u) = \sigma(u)$ for any $\tau \in E' = \text{Aut}_E F$. Therefore, for any $\sigma \in \text{Aut}_K F$, $\tau \in E'$ and $u \in E$, $\sigma^{-1}\tau\sigma(u) = \sigma^{-1}\sigma(u) = u$. Consequently, $\sigma^{-1}\tau\sigma \in E'$ and hence E' is normal in $\text{Aut}_K F$.

Since E is stable, the assignment

$$\sigma \mapsto \sigma|_E$$

defines a group homomorphism $\text{Aut}_K F \rightarrow \text{Aut}_K E$ whose image is clearly the subgroup of all K -automorphisms of E that are extendible to F . Observe that the kernel is $E' = \text{Aut}_E F$ and apply the first homomorphism theorem.

(2) If $\sigma \in \text{Aut}_K F$ and $\tau \in H$, then $\sigma^{-1}\tau\sigma \in H'$ by normality. Therefore, for any $u \in H'$, $\sigma^{-1}\tau\sigma(u) = u$, which implies that $\tau\sigma(u) = \sigma(u)$ for all $\tau \in H$. Thus $\sigma(u) \in H'$ for any $u \in H'$, which means that H' is stable. $\square$

Proposition 3.13. *If E is an intermediate field of the extension F/K such that F/E and E/K are both Galois. Then F is Galois over K if and only if every $\sigma \in \text{Aut}_K E$ is extendible to F*

Proof. Sufficiency. We have

$$\text{Aut}_K F / \text{Aut}_E F \cong \{\sigma \in \text{Aut}_K E : \sigma \text{ is extendible to } F\} = \text{Aut}_K E$$

then

$$|\text{Aut}_K F| = |\text{Aut}_E F| |\text{Aut}_K E| = [F : E] [E : K] = [F : K]$$

It follows from E' is closed and $[K : E'] < \infty$ that K is closed, whence F/K is Galois.

Necessity. Conversely, we have

$$\#\{\sigma \in \text{Aut}_K E : \sigma \text{ is extendible to } F\} = [F : K] / [F : E] = [E : K] = |\text{Aut}_K E|$$

thus $\{\sigma \in \text{Aut}_K E : \sigma \text{ is extendible to } F\} = \text{Aut}_K E$, that is, every $\sigma \in \text{Aut}_K E$ is extendible to F . $\square$

§3.2 Finite Galois correspondence

Theorem 3.14. *Let L/K be a finite extension. The following statements are equivalent:*

1. L/K is Galois.
2. $|\text{Aut}_K L| = [L : K]$.
3. L/K is normal and separable.

Theorem 3.15. *Let L/K be a finite Galois extension, then there is a one-to-one correspondence between the set of all intermediate fields and the set of all subgroups of $\text{Aut}_K L$ given by*

$$X \leftrightarrow X'$$

such that:

1. *the relative dimension of two intermediate fields is equal to the relative index of the corresponding subgroups*

$$\begin{array}{ccccccc} K & \subset & E_1 & \subset & E_2 & \subset & L \\ G & \supset & \text{Aut}_{E_1} L & \supset & \text{Aut}_{E_2} L & \supset & 1 \end{array}$$

then

$$[E_2 : E_1] = [\text{Aut}_{E_1} L : \text{Aut}_{E_2} L]$$

2. L is Galois over every intermediate field E , $E'' = E$

3. E is Galois over $K \Leftrightarrow E' \triangleleft \text{Aut}_K F \Leftrightarrow \{\sigma \in \text{Aut}_K E : \sigma \text{ is extendible to } F\} = \text{Aut}_K E$;
in this case

$$K'/E' \cong \{\sigma \in \text{Aut}_K E : \sigma \text{ is extendible to } F\} = \text{Aut}_K E$$

Theorem 3.16 (Artin). *Let F be a field, G a group of automorphisms of F and K the fixed field of G in F . Then F is Galois over K . If G is finite, then F is a finite dimensional Galois extension of K with Galois group G .*

Proof. In any case G is a subgroup of $\text{Aut}_K F$. If $u \in F - K$, then there must be a $\sigma \in G$ such that $\sigma(u) \neq u$. Therefore, the fixed field of $\text{Aut}_K F$ is K , whence F is Galois over K .

If G is finite, $[F : K] = [1' : G'] \leq [G : 1] = |G|$. Consequently, F is finite dimensional over K , whence $G = G''$ by Lemma 2.10(iii). Since $G' = K$ (and hence $G'' = K'$) by hypothesis, we have $\text{Aut}_K F = K' = G'' = G$. $\square$

§3.3

§3.4 Question

Lemma 3.17. *Let K be a field and a element f/g in $K(x)$ with $f/g \notin K$ and f, g relatively prime in $K[x]$*

Proposition 3.18 ($K(x)/K$). *Let $f/g \in K(x)$ with $f/g \notin K$ and f, g relatively prime in $K[x]$ and consider the extension of K by $K(x)$.*

1. x is algebraic over $K(f/g)$ and $[K(x) : K(f/g)] = \max(\deg f, \deg g)$.
2. If $E \neq K$ is an intermediate field, then $[K(x) : E]$ is finite.
3. The assignment $x \mapsto f/g$ induces a K -homomorphism $\sigma : K(x) \rightarrow K(x)$ such that $\varphi(x)/\psi(x) \mapsto \varphi(f/g)/\psi(f/g)$. σ is a K automorphism of $K(x)$ if and only if $\max(\deg f, \deg g) = 1$.
4. Thus $\text{Aut}_K K(x)$ consists of all those automorphisms induced by the assignment

$$x \mapsto (ax + b)/(cx + d)$$

where $a, b, c, d \in K$ and $ad - bc \neq 0$.

5. If K is an infinite field, then $K(x)$ is Galois over K . If K is finite, then $K(x)$ is not Galois over K .

Proof. (1) x is a root of the nonzero polynomial $\varphi(y) = (f/g)g(y) - f(y) \in K(f/g)[y]$; show that $\deg \varphi = \max \{\deg f, \deg g\}$ and φ is irreducible in $K(f/g)[y]$

Since f/g is transcendental over K , we may for convenience replace $K(f/g)$ by $K(z)$ (z an indeterminate) and consider $\varphi = zg(y) - f(y) \in K(z)[y]$.

Indeed, φ is irreducible in $K(z)[y]$ provided it is irreducible in $K[z][y]$ by Gauss lemma. The truth of this latter condition follows from the fact that φ is linear in z and f, g are relatively prime.

(3) Assume that $\deg g = \max \{\deg f, \deg g\} > 1$. For any $\varphi/\psi \in K(x)$ such that $\varphi, \psi \in K[x], \gcd(\varphi, \psi) = 1$ and $\deg \varphi = m, \deg \psi = n$ ($m > n$), there exist $u, v \in K[x]$ such that

$$u(x)\varphi(x) + v(x)\psi(x) = 1_K$$

and u . Then the image of φ/ψ

$$\varphi(f/g)/\psi(f/g) = \frac{g^k \varphi(f/g)}{g^k \psi(f/g)}$$

where k is sufficiently large that $k > \max \{\deg u + \deg \varphi, \deg v + \deg \psi\}$. Then we have

$$g^k u(f/g)\varphi(f/g) + g^k v(f/g)\psi(f/g) = g^k$$

thus $\gcd(g^m \varphi(f/g), g^m \psi(f/g)) = \gcd(g^m \varphi(f/g), g^{m-n} g^n \psi(f/g))$ is a power of g . Therefore, we have

$$\gcd(g^m \varphi(f/g), g^m \psi(f/g)) = 1$$

If we rewrite

$$\varphi(f/g)/\psi(f/g) = F/G$$

where $F, G \in K[x]$, then $\deg F/G = \max \{\deg F, \deg G\} > 1$, the homomorphism is not surjective.

(5) If K is infinite and $K(x)$ is not Galois over K , then $K(x)$ is finite dimensional over the fixed field E of $\text{Aut}_K K(x)$ by (2). But $\text{Aut}_E K(x) = \text{Aut}_K K(x)$ is infinite (4), which contradicts $[E' : 1] \leq [K(x), E]$.

If K is finite and $K(x)$ is Galois over K , then $\text{Aut}_K K(x)$ would be infinite by Lemma 2.9. But $\text{Aut}_K K(x)$ is finite by (4) $\square$

§4 Galois Groups

§4.1

Definition 4.1. Let K be a field. The **Galois group** of $f \in K[x]$ is the group $\text{Aut}_K F$, where F is a splitting field of f over K .

Theorem 4.2. Let K be a field and $f \in K[x]$ an irreducible polynomial of degree n with Galois group $\text{Aut}_K F$. Then

1. n divides $|\text{Aut}_K F|$
2. $\text{Aut}_K F$ is isomorphic to a transitive subgroup of S_n .

Proof. (1) If $u_1, \dots, u_n$ are the distinct roots of f in some splitting field F ($1 \leq n \leq \deg f$), then every $\sigma \in \text{Aut}_K F$ induces a unique permutation of $\{u_1, \dots, u_n\}$. Consider S_n as the group of all permutations of $\{u_1, \dots, u_n\}$ and verify that the assignment of $\sigma \in \text{Aut}_K F$ to the permutation it induces defines a monomorphism $\text{Aut}_K F \rightarrow S_n$ by

$$\sigma \mapsto \begin{pmatrix} u_1 & u_2 & \cdots & u_n \\ u_{\sigma(1)} & u_{\sigma(2)} & \cdots & u_{\sigma(n)} \end{pmatrix}$$

As for (2), F is Galois over K and $[K(u_1) : K] = n = \deg f$. Therefore, G has a subgroup $K(u_1)' = \text{Aut}_{K(u_1)} F$ of index n by the Fundamental Theorem ($[\text{Aut}_K F : K(u_1)'] = [K(u_1) : K]$), whence n divide $|G|$. For any $i \neq j$ there is a K -isomorphism $\sigma : K(u_i) \cong K(u_j)$ such that $\sigma(u_i) = u_j$. Then σ extends to a K -automorphism of F by Theorem 3.8, whence G is isomorphic to a transitive subgroup of S_n . $\square$

Definition 4.3. Let K be a field with $\text{char } K \neq 2$ and $f \in K[x]$ a polynomial of degree n with n distinct roots $u_1, \dots, u_n$ in some splitting field F of f over K . Let

$$\Delta = \prod_{i < j} (u_i - u_j) = (u_1 - u_2)(u_1 - u_3) \cdots (u_{n-1} - u_n) \in F$$

the **discriminant** of f is the element $D = \Delta^2$.

Proposition 4.4. Let K, f, F and Δ be as in Definition .

(1) The discriminant Δ^2 of f actually lies in K .

(2) For each $\sigma \in \text{Aut}_K F < S_n$, σ is an even [resp. odd] permutation if and only if $\sigma(\Delta) = \Delta$ [resp. $\sigma(\Delta) = -\Delta$].

§5 Finite Fields

Theorem 5.1. Let F be a field and

(1) let P be the intersection of all subfields of F . Then P is a field with no proper subfields. If $\text{char } F = p$ (prime), then $P \cong \mathbb{Z}_p$. If $\text{char } F = 0$, then $P \cong \mathbb{Q}$, the field of rational numbers. The field P is called the **prime subfield** of F .

(2) If F is a finite field, then $\text{char } F = p \neq 0$ for some prime p and $|F| = p^n$ with $n = [F : P] \geq 1$, we have $\mathbb{Z}_p$ -module isomorphism

$$F \cong (\mathbb{Z}_p)^n$$

Theorem 5.2. If F is a field and G is a finite subgroup of $F^\times$, then G is a cyclic group. In particular, the multiplicative group of all nonzero elements of a finite field is cyclic.

Proof. If $G(\neq 1)$ is a finite abelian group, $G \cong Z_{m_1} \oplus Z_{m_2} \oplus \cdots \oplus Z_{m_k}$ where $m_1 > 1$ and $m_1 | m_2 | \cdots | m_k | p^n - 1$.

Since $m_k(\sum Z_{m_i}) = 0$, it follows that every $u \in G$ is a root of the polynomial $x^{m_k} - 1_{F'} \in F[x]$ (G is a multiplicative group). Since this polynomial has at most m_k distinct roots in F , we must have $k = 1$ and $G \cong Z_{m_k}$. $\square$

Corollary 5.3. *If F is a finite field with $\text{char } F = p$. Then*

- (1) $F = \mathbb{Z}_p(u)$ where u is a generator of $F^\times$
- (2)

Lemma 5.4. *If F is a field of characteristic p and $r \geq 1$ is an integer, then the map $\varphi : F \rightarrow F$ given by*

$$u \mapsto u^{p^r}$$

is a $\mathbb{Z}_p$ -monomorphism of fields. If F is finite, then φ is a $\mathbb{Z}_p$ -automorphism of F .

Theorem 5.5. *Let p be a prime and $n \geq 1$ an integer. Then F is a finite field with p^n elements if and only if F is a splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$.*

Proof. It is clear that

$$u^{p^n} - u = 0$$

for all $u \in F$ and all distinct roots of $x^{p^n} - x$ are in F , thus F is a splitting field of $x^{p^n} - x$.

If F is a splitting field of $f = x^{p^n} - x$ over $\mathbb{Z}_p$, then since $\text{char } F = \text{char } \mathbb{Z}_p = p$, $f' = -1$ and f is relatively prime to f' . Therefore f has p^n distinct roots in F . Let $\varphi : u \mapsto u^{p^n}$ be the monomorphism, it is easy to see that $u \in F$ is a root of f if and only if $\varphi(u) = u$. Use this fact to verify that the set E

$$E = \{t \in F : f(t) = 0\} = \langle \varphi \rangle'$$

is a subfield (fixed field of $\langle \varphi \rangle$) of F of order p^n (f splits and has distinct p^n roots), which necessarily contains the prime subfield $\mathbb{Z}_p$. Since F is a splitting field, it is generated over $\mathbb{Z}_p$ by the roots of f (that is, the elements of E). Therefore, $F = \mathbb{Z}_p(E) = E$. $\square$

Corollary 5.6 (Existence and uniqueness of finite fields). *If p is a prime and $n \geq 1$ an integer, then there exists a field with p^n elements. Any two finite fields with the same number of elements are isomorphic.*

Given p and n , a splitting field F of $x^{p^n} - x$ over $\mathbb{Z}_p$ exists by Theorem 3.2 and has order p^n by Proposition 5.6. Since every finite field of order p^n is a splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$ by Proposition 5.6, any two such are isomorphic by Corollary 3.9.

§5.1 Extension over finite fields

Theorem 5.7. *If K is a finite field, then*

- (1) *For any $n \in \mathbb{Z}_{>1}$ there exists a simple extension field $F = K(u)$ such that $[F : K] = n$.*
- (2) *Any two n -dimensional extension fields of K are K -isomorphic.*
- (3) *For any $n \geq 1$ an integer, there exists a minimal polynomial of degree n in $K[x]$.*

Proof. (1) Given K of order p^r let F be a splitting field of

$$f = x^{p^{rn}} - x$$

over K . By Proposition 5.6 every $u \in K$ satisfies $u^{p^r} = u$ and it follows inductively that $u^{p^{rn}} = u$ for all $u \in K$. Therefore, F is actually a splitting field of f over $\mathbb{Z}_p$. Since F consists of precisely the p^{nr} distinct roots of f , we have

$$p^{nr} = |F| = |K|^{[F:K]} = (p^r)^{[F:K]}$$

whence $[F : K] = n$. Corollary 5.4 implies that F is a simple extension of $\mathbb{Z}_p$, hence of K .

(2) Uniqueness. If F_1 is another extension field of K with $[F_1 : K] = n$, then $[F_1 : \mathbb{Z}_p] = n[K : \mathbb{Z}_p] = nr$, whence $|F_1| = p^{nr}$. By Proposition 5.6 F_1 is a splitting field of $x^{p^{nr}} - x$ over $\mathbb{Z}_p$ and hence over K . Consequently, F and F_1 are K -isomorphic, hence are isomorphic. $\square$

Theorem 5.8. *If F is a finite dimensional extension field of a finite field K (It equivalent that $\mathbb{Z}_p \subset K \subset F$ are finite extension with $[F : \mathbb{Z}_p] = n$, $[K : \mathbb{Z}_p] = r$), then*

- (1) $r \mid n$
- (2) F is Galois over K .
- (3) The Galois group $\text{Aut}_K F = \langle \varphi^r \rangle$ is cyclic.

Proof. (1) Consider

$$[F : K][K : \mathbb{Z}_p] = [F : \mathbb{Z}_p]$$

that is, $[F : K]r = n$

(2) F is a splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$. It follows from all roots of $x^{p^n} - x$ are distinct that F is Galois over $\mathbb{Z}_p$, hence over K .

(3) The map $\varphi : F \rightarrow F$ given by $u \mapsto u^p$ is a $\mathbb{Z}_p$ -automorphism of F with order n . Since $|\text{Aut}_{\mathbb{Z}_p} F| = [F : \mathbb{Z}_p] = n$ by the Fundamental Theorem, $\text{Aut}_{\mathbb{Z}_p} F$ must be the cyclic group generated by φ .

Since $\text{Aut}_K F$ is a subgroup of $\text{Aut}_{\mathbb{Z}_p} F$, $\text{Aut}_K F$ is also cyclic of order $[F : K] = n/r$. On the other hand, $\varphi^r : u \mapsto u^{p^r}$, automorphism of F , fix K and φ^r is of order n/r . Therefore, we have

$$\text{Aut}_K F = \langle \varphi^r \rangle$$

$\square$